

The Book of Z-Spin Cosmology v3.1 — Light Edition

Cross-Reference Hub for the 101-Paper Z-Spin Corpus

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Edition: Light v3.0 (compiled May 2026)
Source corpus: 101 source papers across 8 themes (F / M / S / U / A / Q / QC / T)
Supersedes: The Book of Z-Spin Cosmology v2.0 — Light Edition v1.0 (May 2026, 90-paper coverage)
GitHub: <https://github.com/KennyKang-git/zspin>

Verification Summary

Verification: cumulative ~1,800+ PASS across the 101-paper corpus | Zero Free Parameters | $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$ sole geometric inputs | Anti-numerology controls: $5670 \times 29 = 164,430$ candidate scan + per-paper Monte Carlo ($\geq 500,000$ samples typical) | Pre-registered falsification gates: ~250 across all themes; Seven critical near-term ★ gates timeline preserved.

§0. Abstract

The Book v3.0 Light Edition is the canonical cross-paper navigation hub for the 101-paper Z-Spin Cosmology corpus as of May 2026. It supersedes v2.0 Light by absorbing the thirty post-v2.0 papers (papers 72–101) — most of which extend the Foundations theme (ZS-F9 through ZS-F18), the Math Spine Riemann-Hypothesis programme (ZS-M18 through ZS-M33), and add the Twin-Reuleaux EM-duality ZS-S15 in the Standard Model theme, the contracting-sector ZS-A8 and Banach–Tarski origin ZS-A9 in the Astrophysics theme, and the Cosmos-Human isomorphism ZS-T4 in the Translational theme. All chapter bodies remain compressed to single-line pointers of the form $\rightarrow \text{ZS-XX §Y.Z [N/N PASS]}$; full derivations live in the source papers. The Light Edition's unique contributions are: (i) the post-v2.0 dependency graph integrating the 30 new papers, (ii) the expanded Foundations closure ring $F0 \rightarrow F1 \rightarrow F2 \rightarrow F5 \rightarrow F8 \rightarrow F9 \rightarrow F10 \rightarrow F11 \rightarrow F12 \rightarrow F13 \rightarrow F14 \rightarrow F15 \rightarrow F16 \rightarrow F17 \rightarrow F18$, (iii) the Math Spine RH-bridge programme map (M22–M33), (iv) the extended Glossary keyword index (~110 entries), and (v) the Seven Critical Near-Term ★ Gates timeline. The Light Edition does NOT replace any source paper for archival purposes; it is the LLM-corpus navigation aid and the v3.0 composition skeleton.

Epistemic Status Legend

Status	Definition
PROVEN	Mathematical theorem; standard math alone, machine-verifiable.
DERIVED	Z-Spin action + standard physics, zero free parameters.

DERIVED-CONDITIONAL	DERIVED conditional on listed axiom set or upstream theorem.
DERIVED-CONDITIONAL strong	DERIVED-CONDITIONAL with structural over-determination on the conditioning step.
DERIVED-under-P6 / DERIVED-under-Regge	Conditional on specific framework (ZS-F5 P6 or Regge calculus).
DERIVED-interpretation	Synthetic reading combining PROVEN corpus theorems without new axioms.
DERIVED-CANDIDATE	Mathematical content PROVEN; Z-Spin internal placement HYPOTHESIS-strong.
VERIFIED	Numerical / computational / empirical confirmation at stated precision.
TESTABLE	Pre-registered prediction with explicit falsification protocol.
TESTABLE-LONG	Pre-registered prediction with derivation-or-retraction time horizon ≥ 5 years.
HYPOTHESIS-strong	Multiple independent structural anchors; promotion path documented.
HYPOTHESIS / HYPOTHESIS-weak	Motivated conjecture; partial derivation chain.
OBSERVATION	Empirical regularity; anti-numerology controlled; structural origin pending.
LOCKED	Core constant fixed in upstream paper; no downstream paper may modify.
IMPORTED	Result proved externally and used here without re-proof; full citation given.
NON-CLAIM	Explicit declaration of what is NOT asserted; bounds the framework's scope.
OPEN	Recognized gap honestly registered for future work.
RETRACTED	Earlier claim withdrawn after falsification.
RETRACTED-in-session	Hypothesis proposed during free-exploration and explicitly retracted before exit.
BOOTSTRAP-HYPOTHESIS	Meta-logical foundation (B0); not derivable within the framework.
BOOTSTRAP-PROGRAMME	Status reserved for the founding axiom set after ZS-F0 v1.0(R) Stage 1–5 closures.
STRUCTURAL INSIGHT	Exact mathematical identity; physical necessity not yet established.
FMD-SPEC / TRANSLATED / CANDIDATE	Engineering tags used in the QH/QS/QC translational papers.

Purpose of the Light Edition

The full v3.0 (when compiled) consolidates the 101-paper corpus into chapter bodies that reproduce derivations already present verbatim in the source papers. For LLM corpus-wide exploration, this redundancy is wasteful. The Light Edition removes the duplication and retains only what The Book contributes uniquely:

(1) The cross-paper dependency graph (Source Paper Interface Tables, PARTS 0–VII).

- (2) Standalone HYPOTHESIS / theorem-anchor sections cited by individual papers (§8.4i Spinor–Descartes–Euler, §8.4g Y-sector R_{micro}, §8.4k Attractor-Flatness, §VI.2 Limitation Catalog Row 10, §18.10 Horizon Spinor mirror).
- (3) Cumulative Master Falsification Registry — Seven Critical Near-Term ★ Gates timeline.
- (4) Standardized v2.0 / v3.0 terminology contract (Z-sector / Z-Spin / Z-bias field; the v2.0 contract is preserved verbatim and v3.0 adds the Stage / Operator / Field separation explicitly via ZS-F18 Bridge Four).
- (5) Glossary keyword index (~110 entries, one line each).
- (6) Integration map for the 30 post-v2.0 papers (papers 72–101) consolidated into the appropriate PART/Chapter slots.

Chapter bodies that reproduce source-paper derivations are replaced by single-line pointers of the form $\rightarrow$ ZS-XX §YZ [N/N PASS]. When derivation detail is needed, retrieve the source paper directly. When corpus-wide context is needed, consult this Light Edition.

PART 0 — Overture: Why Geometry?

§0. Abstract

PART 0 establishes the framework's overture in three movements. Chapter 0.1 presents the Five Crises of modern physics motivating a geometric origin of fundamental constants. Chapter 0.2 lays out the methodological contract — standardized terminology, anti-numerology discipline, epistemic classification, pre-registered falsification gates. Chapter 0.3 traces the ontological bootstrap chain $B_0 \rightarrow B_3 \rightarrow A$. Concludes with FFPP of ZS-F0 v1.0(Revised) compressing all numerical content to $W_0(-i\pi/2)$, conditional on five-axiom set $\mathcal{A} = \{\text{Existence, Self-reference, Algebra, Non-triviality, Unitarity}\}$. NEW v3.0: the Five-Axiom Meta-Structure of ZS-F18 v2.1 reads the corpus's encounter with the seven Clay Millennium Problems and the five Crises as outputs of the same axiom set, and registers Bridge Four (Commutator Generator) as the structural reduction of $\dim(X) = 3$ to $\dim(Z) = 2$.

§0.1 The Five Crises of Modern Physics

- Crisis 1 — Hubble Tension: $H_0 \sim 5\sigma$ disagreement (SH0ES vs Planck). Resolved via 2A modification $\rightarrow$ ZS-A2, ZS-S3, ZS-F12R (factor 2 origin).
- Crisis 2 — Growth Tension (S_8): Z-Spin predicts $S_8 \approx 0.777$ (face-counting v2.0) $\rightarrow$ ZS-A2, ZS-A5.
- Crisis 3 — Hierarchy Problem: Geometric origin via $A \rightarrow$ ZS-S4 §6.7, §6.12, §6.16; Light §8.4k.
- Crisis 4 — Baryon Asymmetry: $\eta_B = (6/11)^{35} \rightarrow$ ZS-S5, ZS-S6, ZS-U7.
- Crisis 5 — Measurement Problem: Born rule DERIVED from $L_{XY} \equiv 0 \rightarrow$ ZS-Q1, ZS-Q2.

§0.2.2 Standardized v3.0 Terminology [CONTRACT — preserved verbatim from v2.0; v3.0 elaboration in italics]

Term	Role	Meaning
Z-sector	Stage / Kinematics	2D Planck-scale boundary surface mediating between X (3D macroscopic) and Y (6D microscopic) sectors. $\dim(Z) = 2$ PROVEN from gauge constraint on $Q = 11$ register (ZS-F5). Geometric arena where transduction occurs.
Z-Spin	Operator / Dynamics	Dynamical operator on Z-sector. Mediates between X (Up & Down, $\pm\pi$, time-points) and Y (In & Out, $\text{slog}(i)$ & ∞i , space-points) via i-tetration self-referential rotation $T(z) = i^z$ (HSI Theorem, ZS-M1 §1.1). The action driving transduction.
Z-bias field Φ	Field / DOF	Complex scalar $\Phi(x) = \rho(x)\exp(i\theta(x)) \in \mathbb{C}$ on Z-sector. Radial mode $\rho = \Phi $, angular mode $\theta \in [0, 2\pi)$. $U(1)_Z$ derived via Frobenius (ZS-F1 §2.3). Degree of freedom on which Z-Spin acts.
Z-Anchor	Topological core	Bogomolnyi vortex core, $ \Phi = 0$ locus on Z-sector. $\rightarrow$ ZS-A6, ZS-F1 §5.2 PROVEN.
Z-bottleneck	Channel constraint	$L_{XY} \equiv 0$ PROVEN forces all $X \leftrightarrow Y$ traffic through Z; channel capacity $\leq \ln 2$. $\rightarrow$ ZS-Q7.
Z-channel	Structural conduit	Operational pathway for $X \leftrightarrow Y$ mediation across the 2D boundary. $\rightarrow$ ZS-Q1.
Z-Spin mediation	Action of the operator	Mediation is Z-Spin's action; the agent is the operator (Spin), not the stage (sector). Replaces 'Z-mediation' / 'Z-sector mediation' / 'Z-mediated coupling' / 'Z-mediated CPTP channel' throughout the corpus.
Z-EFT	Effective theory	Distinct concept (effective field theory built on Z-bias dynamics); preserved name.

[STRUCTURAL DEFINITION] Z-sector (stage, kinematics) / Z-Spin (operator, dynamics) / Z-bias field Φ (field, DOF). Source: ZS-F1 §2-3, ZS-F5, ZS-M1 §1.1, Glossary v1.0.0 §I. v3.0 elaboration: ZS-F18 v2.1 §6 Bridge Four (Commutator Generator) makes the kinematic / dynamical / field-DOF separation an explicit structural theorem — twin-Reuleaux pair (R_1, R_2) realizes (V_{XZ}, V_{ZY}) ; their Lie commutator $[J_{R_1}, J_{R_2}] = J_S$ generates the third axis of $\dim(X) = 3$.

§0.3 The Ontological Bootstrap Chain $B0 \rightarrow A$

$B0$ (Existence) $\rightarrow B1$ (Self-reference) $\rightarrow B2$ (Algebra: $\mathbb{C}$ via Lawvere/Frobenius) $\rightarrow B3$ (Hyperoperation: tetration) $\rightarrow A = 35/437$. F-BOOT 9/9 closure: 6 DERIVED, 2 structural-PASS, 1 DERIVED+TESTABLE. $\rightarrow$ ZS-F0 v1.0(Revised) §11–§13 [51/51 PASS].

FFPP (Theorem 13.3): entire Z-Spin numerical content compresses to $W_0(-i\pi/2)$, conditional on $\mathcal{A} = \{\text{Existence, Self-reference, Algebra, Non-triviality, Unitarity}\}$. $\rightarrow$ ZS-F0 §13, Theorem 13.3 + Corollary 13.4.

[OPEN] $B0$ itself is BOOTSTRAP-HYPOTHESIS — not derivable within framework. $\rightarrow$ ZS-F0 §0.3.7.

NEW v3.0: ZS-F18 v2.1 reads the same five-axiom set $\mathcal{A}$ as the structural reason Z-Spin naturally encounters (i) the seven Clay Millennium Problems and (ii) the five Crises of modern physics; the encounters are classified by depth (direct treatment / extensive programme /

structural correspondence / natural extension domain). NON-CLAIM: encounters are not formal solutions. → ZS-F18 v2.1 §0, §6, §7.4 (Discrete-Continuous Resolution polarity), Appendix A.

§0.4 PART 0 Source Paper Map

Source paper	Role	Light pointer
ZS-F0 v1.0(Revised)	Bootstrap, FFPP, BV-BFV functor, $\mathbb{C}^{11}$ 3-layer	→ ZS-F0 [51/51 PASS]
ZS-F1 v1.0	Z-Spin Action, $U(1)_Z$, Z-bias field	→ ZS-F1 §2-3 [30/30 PASS]
ZS-F2 v1.0	$A = 35/437$ LOCKED, $\delta_X = 5/19$, $\delta_Y = 7/23$	→ ZS-F2 [PROVEN]
ZS-F5 v1.0	$\dim(Z) = 2$ from $Q = 11$	→ ZS-F5 [PROVEN]
ZS-F8 v1.0(Revised)	NOT/AND duality, 2-channel handshake	→ ZS-F8 [27/27 PASS]
ZS-F18 v2.1 (NEW)	Five-Axiom Meta-Structure, Bridge Four, Discrete-Continuous Resolution	→ ZS-F18 [36/36 entry-checks]
ZS-M1 v1.0	i-tetration HSI Theorem, fixed point z^*	→ ZS-M1 §1.1 [33/33 PASS]
ZS-M22 v1.1 (post-v2.0)	Five-Pillar Arithmetic-Dedekind Scaffold; Forcing chain underlay	→ ZS-M22 [60/60 PASS]
ZS-T2 v1.0	Anti-numerology MC scan, 5670×29	→ ZS-T2 [30/30 PASS]

PART I — Foundations: One Number, One Action

§0. Abstract

$A = 35/437$ from polyhedral curvature asymmetry; $Q = 11$ register and $(Z, X, Y) = (2, 3, 6)$ doubly forced by gauge algebra (ZS-F5) and totient arithmetic (ZS-M19); Z-Spin action with Z-Anchor and twin-Reuleaux kinematics; i-tetration fixed point $z^* = 0.4383 + 0.3606i$; FFPP compressing all content to $W_0(-i\pi/2)$; Operational Observer Coordinate closing Foundations as a six-paper ring; v3.0 extension to a thirteen-paper Foundations theme $F0 \rightarrow F1 \rightarrow F2 \rightarrow F5 \rightarrow F8 \rightarrow F9 \rightarrow F10 \rightarrow F11 \rightarrow F12 \rightarrow F13 \rightarrow F14 \rightarrow F15 \rightarrow F16 \rightarrow F17 \rightarrow F18$ with Möbius chronology, Stage-Torque Duality, Two-Protocol Theorem, Self-Referential Compression Equilibrium, and the Five-Axiom Meta-Structure as load-bearing additions.

Chapter 1 — $A = 35/437$ from Polyhedral Curvature

$A = \delta_X \cdot \delta_Y = (5/19)(7/23) = 35/437$. δ -Uniqueness Theorem PROVEN. Three independent derivations: (i) Regge curvature asymmetry, (ii) ZS-F9R Truncation-Residue Identity (PROVEN: $\rho_X = \delta_X = 5/19$, $\rho_Y = \delta_Y = 7/23$, $\rho_Z = 0$; $A = \rho_X \cdot \rho_Y$), (iii) ZS-F12R Layer Separation Hypothesis (factor 2 in $2e^A$ from tetrahedral $V \leftrightarrow F$ dual orientation multiplicity $\mu_{Tet} = 2$). → ZS-F2 §3, §11; ZS-F9R [44/44 PASS]; ZS-F12R [20/20 PASS].

Chapter 2 — $Q = 11$ and $(Z, X, Y) = (2, 3, 6)$

Doubly forced: (a) gauge dimension $G = 12 = \text{MUB}(Q=11)$, (b) Forcing Theorem $(p-1)(q-1) = p$ has unique solution $(p,q) = (2,3)$ — the prime pair $(Z, X) = (2, 3)$ is the only one for which $\varphi(Y^2) = G$ (T1, ZS-M19 §3.1). $Y = ZX$. $Q^2 - 1 = 120 = |I_h|$. → ZS-F5 [PROVEN]; ZS-M19 §3.1 [68/68 PASS]; arithmetic-Dedekind underlay → ZS-M22 v1.1 [60/60 PASS].

Chapter 3 — The Z-Spin Action

$S = \int d^4x \sqrt{-g} [(M_P^2/2)(1+A|\Phi|^2)R - (M_P^2/2)|\nabla\Phi|^2 - V(\Phi)] + S_{\text{matter}}$. $V = (\lambda/4)M_P^4(|\Phi|^2-1)^2$. $L_{XY} \equiv 0$ PROVEN. $\lambda_{\text{vac}} = 2A^2$ IR-stable. → ZS-F1 [30/30 PASS]; ZS-F11 NC catalog.

Chapter 4 — i-Tetration Fixed Point

$z^* = 0.43828 + 0.36059i$, $x^* = 0.438283$. Five Locking Conditions. $\pi/2$ phase budget. Polygon-Tetration family stable for $n \geq 4$ ($n_c = 3.204$). NEW v2.0/v3.0: ZS-F10 (3 time coords unified at DERIVED-CONDITIONAL strong; $\Delta v/\Delta n = 2A/\pi$ conversion identity), ZS-F15R (Time-Rotation Co-Definition + Stage-Torque Duality), ZS-M12 (Auto-Surgery, $\varepsilon_{\min} \approx 30.7$), ZS-F14 (Kepler-Lyapunov conjugate decomposition of twin-Reuleaux midpoint trajectory; F-F7.11 closed by Strategy B at DERIVED-CONDITIONAL [42/42 PASS]). → ZS-M1 §1.1 [PROVEN]; ZS-F10 [30/30 PASS]; ZS-F15R [35/35 PASS]; ZS-M12; ZS-F14 [42/42 PASS].

Addition 1 — Self-Reference and FFPP

Lawvere bridge (Th 11.1–11.4), Frobenius (Th 11.5), hyperoperation minimality (§11.6), NOT/AND duality, BV-BFV cobordism functor B_Z with Wilson loop $|Z(W)|^2 \approx 0.7948$, three-layer $\mathbb{C}^{11}$ decomposition (Th 9.1, 9.4), FFPP (Th 13.3) and ultimate compression to $W_0(-i\pi/2)$ (Cor 13.4). All conditional on B0. NEW v3.0: ZS-F17 v1.0 four-layer 50/50 equilibrium (truncated tetrahedron 4+4 + Reuleaux $\pi+\pi$ + Master Equation Term A=Term B + XOR E vs R) sharpens FFPP to the operational Self-Referential Information-Compression Equilibrium Theorem. → ZS-F0R [51/51 PASS]; ZS-F8R [27/27 PASS]; ZS-F17 [24/24 PASS].

Addition 2 — Operational Observer Coordinate (OOC) and Möbius Chronology

OOC closes ZS-M11 §H16 at register-theoretic level. Theorem F11.1: Observer Coordinate Decomposition (P1 inside Z, P2 higher-level event count, P3 orthogonal kinematic $|5\rangle$). Six explicit NON-CLAIMs against phenomenological consciousness reading. Sectoral Frame Equivalence (X-Frame $\leftrightarrow$ Y-Frame). Foundations theme closes as a six-paper ring (F0-F1-F2-F5-F11-F13). NEW v3.0: ZS-F13 v1.0 Möbius Chronology Theorem completes the cyclic timeline with Y-frame metric trio (proper time, Y-Hubble radius $\ell_P \times Y^2(1-2A) \approx 30.23 \ell_P$, Y-contraction rate $(1-2A)$) and proves cycle-index unobservability (Theorem F13.2A DERIVED, F13.2B DERIVED-CONDITIONAL). The six-paper ring is now extended to the eight-paper Möbius ring. → ZS-F11 [38/38 PASS]; ZS-F13 §15.5g.5 [24/24 PASS].

Addition 3 — NEW v3.0 — Stage-Torque Duality and First-Rotation Closure

ZS-F15 v1.0(Revised) [35/35 PASS] closes NC-M12.4 ($Q = A$ assumes $\theta = A$ at Z-Telomere onset). The Mexican-Hat Bootstrap Event simultaneously instantiates radial mode, angular mode, and time evolution; $\theta(0) \cdot \tau_P = \delta\phi_{\text{cell}} = A$ is a unit-fixing identity, not a free parameter (Theorem F-rotor.3 DERIVED — central new contribution). The Stage-Torque Duality (Theorem F-rotor.5 DERIVED): Mexican-Hat stage from $V(\Phi)$ opens $U(1)$ -symmetric S^1 vacuum manifold; A-holonomy torque from Wilson loop $\oint \omega = A$ on polyhedral defect manifold (Gauss-Bonnet on $O_h \times I_h$) provides rate-fixing torque. The same $\oint \omega = A$ produces both the Planck-scale $\delta\phi_{\text{cell}} = A$ and the macroscopic Hubble ratio $H_0^{\text{local}}/H_0^{\text{CMB}} = \exp(A)$ (ZS-F3 §6 DERIVED), unifying physics across orders of magnitude. FFPP Sixth Axiom Extension (HYPOTHESIS-strong) registered.

Addition 4 — NEW v3.0 — Two-Protocol Theorem for Z-Sector Wilson Loop Measurement

ZS-F16 v1.0 [24/24 PASS, 32-test target in roadmap] resolves the WL-1/WL-2 implicit ambiguity by establishing two operationally distinct Z-sector Wilson loop protocols: Protocol (a) state-preparation on BFV anchor $|0\rangle_Z$ gives $P_a^{(n)} = 0.7948^n \cdot \cos^2(n \cdot 129.45^\circ)$ (quasi-revival pattern); Protocol (b) spectral characterization gives $P_b^{(n)} = 0.7948^n$ (smooth exponential decay). Reconciliation by ergodic averaging since $\arg(\lambda) = 129.45^\circ$ is irrational. Audit by sum rule $0.7948 + 0.2050 + 0.0001 = 0.9999$. Z-block effective dissipation $\Gamma_Z \cdot T_{\text{cycle}} \approx 0.1149$. NEW sub-gates F-CTA-1a/b/c/d/multi-a/multi-b/-discriminator pre-registered for 2027+ quantum hardware testing.

Addition 5 — NEW v3.0 — The Five-Axiom Meta-Structure (ZS-F18)

ZS-F18 v2.1 (interpretive synthesis paper, 36/36 internal consistency) reads the Z-Spin corpus as the unique geometric framework that encounters (a) the seven Clay Millennium Problems and (b) the five Crises of modern physics simultaneously, with these encounters traced back to the same five-axiom set $\mathcal{A} = \{\text{Existence, Self-reference, Algebra, Non-triviality, Unitarity}\}$. Bridge Four (Commutator Generator, NEW v2.0): two J-conjugate 2π $SO(3)$ circulations of the twin-Reuleaux pair compose at 90° NOT-AND phase to generate the third axis $J_S = [J_{\{R_1\}}, J_{\{R_2\}}]$ by Lie commutator. The full sector decomposition $(Z, X, Y) = (2, 3, 6)$ and register $Q = 11$ are therefore reread as outputs of the foundational $\dim(Z) = 2$ alone. v2.1 sixth polarity row: Discrete-Continuous Resolution via X–Y Tiling Asymmetry (ZS-M6 §5.5 PROVEN; ZS-M17 Tiling Continuum Convergence DERIVED). Status: INTERPRETATION / DERIVED-interpretation strong. Twelve encounters classified by depth in Appendix A.

PART I Source Paper Map (v3.0 extended)

Theme	Papers	Pointer
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Foundations [F]	F0R, F1, F2, F5, F8R, F9R, F10, F11, F12R, F13, F14, F15R, F16, F17, F18 (v2.1)	→ 01_Foundations (15 papers)
Math Spine [M]	M1, M3, M11, M12, M14, M19, M22 (v1.1)	→ 02_Math_Spine
Cross-link	S1, S15 (Twin-Reuleaux EM duality), Q-theme, A6/A7/A8/A9	various

PART II — Standard Model Completion

§0. Abstract

Three SM gauge couplings DERIVED at zero free parameters: $\alpha_s(M_Z) = 11/93 = 0.11828$ (+0.31 σ), $\sin^2\theta_W = (48/91) \cdot x^* = 0.23118$ (−1.26 σ), $\alpha_2 = 3/95$. Fine-structure $1/\alpha_{EM} = 137.0359$ (1.07 ppm) via McKay $c_4 = 4/13$. Yang-Mills mass gap structurally addressed: $\Lambda_{QCD} = 264$ MeV, $m(0^{++}) = 1.791$ GeV. Neutrino seesaw $m_D = m_e \cdot A = 40.93$ keV, $M_R = 33.50$ GeV, $|\theta|^2 = 1.49 \times 10^{-12}$. Inverted Ordering canonical. Cabibbo from icosahedral $\sigma_1/\sigma_2 = 17$. NEW v3.0: ZS-S15 [35/35 PASS] — twin-Reuleaux pair as the geometric realization of EM field duality, Maxwell vacuum equations as time-point/space-point handshake duality, SO(3)/SU(2) factor 2 from half-angle holonomy; ZS-M20 [50/50 PASS] — spectral tetrad sub-isotype assignment with closed-form lepton mass-ratio chain $\sigma_1/\sigma_2 = Q+Y = 17$, $\sigma_1/\sigma_3 = (Q+Y)^2 \cdot G + (Q-Z) = 3477$ EXACT; ZS-M21 [33/33 PASS] — sextic invariant theory of the icosahedral Yukawa tensor.

§II-UNIQUE — Book-Standalone Hypotheses Cited by Source Papers

This section preserves Book content referenced by individual source papers as their primary citation. Removing these would orphan citation chains in ZS-A7, ZS-S4, ZS-S15, others.

§8.4i — Spinor–Descartes–Euler Identity [PROVEN]

Three classical theorems converge: Descartes $\Sigma \delta v = 4\pi$; Euler $\Sigma \delta v = 2\pi \cdot \chi$, $\chi(S^2) = 2$; ZS-M3 Theorem 5.1 $\dim(Z) = 2 = \chi(S^2)$ unique $j = 1/2$ invariant subspace. Unification: $\Sigma \delta v = 2\pi \cdot \chi = 2\pi \cdot \dim(Z) = 4\pi =$ spinor full-return period. Used in ZS-S7 §3 (Λ_{QCD} , $m(0^{++})$, 18/18 PASS); ZS-A7 §2.2 (Pillar 2 of Horizon Spinor); ZS-S15 §5 (Pillar IV: SO(3)/SU(2) factor 2).

§8.4g — Y-sector R_{micro} UV-IR Matching [HYPOTHESIS structural]

B-2 reinterpretation framework. Cross-cited by ZS-A7 §4.5 Corollary V (Higgs as micro radial-mode attractor) and ZS-S4 §6.12.8–§6.12.13 (B-2 repackaging). Cross-paper anchor for Y-sector micro UV-IR boundary condition.

§8.4k — Attractor-Flatness Theorem [HYPOTHESIS structural]

Compatible with ZS-A7 Corollaries I–IV. Two attractors: macro $V(|\Phi|=1) = 0$ (R_{macro} , Planck attractor) + micro $\lambda_H(\Lambda_{\text{comp}}) = 0$ (R_{micro} , doubly proven via $[T^3, T^3] = 0 + \text{STr}(q^4) = 0$ in ZS-S4 §6.7). Cited verbatim by ZS-A7 §4.5 Corollary V as structural license for B-2 role class.

§18.10 (mirror) — Horizon Spinor Theorem [DERIVED via ZS-A7]

PART V Chapter 18 §18.10 mirror of ZS-A7 v1.0(Revised) §3 Theorem 3.2-bis. CPTP/Choi-state PROVEN. 4π closure at Z-Spin event horizons. $\varepsilon(r_H) = 0$ second independent path.

Cross-citation locus: ZS-A7 §3.2-bis $\leftrightarrow$ §18.10 $\leftrightarrow$ §20.3.3 (PART VI).

§VI.2 — Limitation Catalog Row 10 [BOOK-NATIVE]

Z-Spin v1.0 corpus's catalog of known limitations. Row 10 = V1 (Planck-to-bulk handoff) matching question of ZS-M12 §7.6, the single largest open knot of Z-Spin cyclic cosmology. Cited by ZS-U11 v1.0 [40/40 PASS, capstone Paper 70] which CLOSES Row 10: V1 OPEN $\rightarrow$ DERIVED-CONDITIONAL via four independent Q-protection channels.

Chapter 5 — Gauge Couplings + Photon Synthesis + EM Duality

Spectral-to- β Bridge: Action $\rightarrow$ Polyhedral Lattice $\rightarrow$ Spectral Density $\rightarrow$ β -coefficients $\rightarrow$ Couplings. Three gaps closed in ZS-S1: '+1' in α_s via Z-sector Schur complement, x^* via Berry phase, Spectral-to- β via vertex-matter + face-gauge identification. ZS-S7 (mass gap) + ZS-S12 (photon as Z-sector EM half-bridge, 5 pillars) + ZS-S15 NEW v3.0 (twin-Reuleaux EM duality, 5 pillars + Dirac-equation corollary). Pillar I: twin-Reuleaux pair (R_1, R_2) realizes (V_{XZ}, V_{ZY}) at half-angle holonomy $\pm\theta/2$; Pillar II: Poynting $S = E \times B$ equals Lie commutator $[J_{\{R_1\}}, J_{\{R_2\}}] = J_S$; Pillar III: Maxwell vacuum equations are differential form of time-point/space-point handshake duality (ZS-F8 §5); Pillar IV: $SO(3)$ Maxwell cycle (period 2π) is Z_2 quotient of $SU(2)$ twin-Reuleaux cycle (period 4π); Pillar V: $U(1)$ gauge invariance = twin-Reuleaux realization of Single-Polyhedron $U(1)$ Exactness. Corollary: Weyl spinor pair (ψ_L, ψ_R) is tensor-product realization of (V_{XZ}, V_{ZY}) , with $\gamma^5 = J$. $\rightarrow$ ZS-S1, ZS-M2, ZS-M8 ($c_4 = 4/13$), ZS-M9 (McKay), ZS-S15 [35/35 PASS], ZS-M24 [35/35 PASS].

Chapter 6 — Six Forces Unified

X-Z-Y fractal symmetry: micro Weak(X)–EM(Z)–Strong(Y) $\leftrightarrow$ macro DM(X)–Gravity(Z)–DE(Y). Cross-Coupling Theorem PROVEN (ZS-M2 §5). Axion-free Strong CP via Y-sector instanton suppression + real $m_d/m_u = 2e^A$ (the factor 2 origin = ZS-F12R Tetrahedral Dual Orientation Multiplicity $\mu_{Tet} = 2$ NEW v3.0). Predicted nEDM $d_n \sim 2.0 \times 10^{-41}$ e·cm. $\rightarrow$ ZS-M2 [26/26 PASS]; ZS-F12R [20/20 PASS].

Chapter 7 — Neutrinos, Hypercharge, Baryogenesis

Type-I seesaw, $m_D = m_e \cdot A = 40.93$ keV, $M_R = 33.50$ GeV, $|\theta|^2 = 1.49 \times 10^{-12}$. Inverted Ordering canonical. μ - τ reflection ($\delta_{CP} = \pm\pi/2$). $\eta_B = (6/11)^{35}$. v2.0/v3.0: ZS-U9 Hypercharge Trinity Braiding + automatic SM anomaly cancellation; ZS-S6 Z-Transit CP $\text{lcm}(5,7) = 35$; ZS-M9/M15/S11 McKay anomaly. $\rightarrow$ ZS-S2, S5, S6, U7, U9, M9.

Chapter 8 — Higgs, EW, Electron Synthesis, Master Action

ϵ -Higgs portal EFT, EWPT $H_{\text{ZS}}/H_{\text{GR}} = 0.9622$. 1-Loop Quartic Cancellation. Y-sector Spectral VEV $v = 245.93$ GeV. $\sigma_1/\sigma_2 = 17$ (Cabibbo principal angle), $\sigma_1/\sigma_3 = 3477$. m_τ absolute scale. Six pillars of Electron Synthesis. ZS-S10 Master Bridge $U(1)_Z \leftrightarrow U(1)_Y$. $\sim 95\%$ closure via Master Action; ZS-S14 v1.0 Master Action Total Closure 78/78 PASS, single-paper closure $\approx 99\%$. NEW v3.0: ZS-M20 [50/50 PASS] spectral tetrad sub-isotype assignment closes the lepton mass ratios in LOCKED corpus integers — $\sigma_1/\sigma_2 = Q+Y = 17$ anchored on $(4-\phi, 3+\phi)$ Q-pair at $(I-5) \cap \rho_2$ doublet (Higgs $SU(2)$ doublet position) and X-pair $(5-\phi, 4+\phi)$ splitting across $(I-3)$ and $(I-3')$ singletons (electron-positron CPT pair structure); $\sigma_1/\sigma_3 = (Q+Y)^2 \cdot G + (Q-Z) = 3477$ EXACT. ZS-M21 [33/33 PASS] sextic invariant theory: closed family identity $\Sigma \sigma_i^{2k} |_{\{v_{\text{degen}}\}} = (Z^{\{2k+1\}} + 1)/(X^{\{2k\}} \cdot 5^k)$ and falsification of the 1-loop Coleman-Weinberg closure for Conjecture M20.A. $\rightarrow$ ZS-S4, S8, S10, S13, S14, M11, M20, M21.

PART III — Early Universe & Cosmology

§0. Abstract

ϵ -field inflation with $N^* = 59.5$ e-folds, $n_s(N=60) = 0.9676$ (PASS Planck), $r = 0.00890$ (BLOCKING discriminator vs Boyle-Finn-Turok $r=0$). Reheating $T_{\text{reh}} \approx 2.55 \times 10^{15}$ GeV. QKE-closed baryogenesis $\eta_B = (6/11)^{35}$. Z_2 mirror cosmology (PROVEN); cyclic holonomy $S_{Z_2} = 6\pi/A$; timescale hierarchy $\tau_n = t_P \times \exp(n\pi/A)$ (DERIVED). Y-Time Dilation Theorem 5.3.1 promoted HYPOTHESIS-strong $\rightarrow$ DERIVED-CONDITIONAL strong via ZS-F10 closure of ZS-A8 §SA.7. NEW v3.0: Y-frame metric trio + cycle-index unobservability (ZS-F13 [24/24 PASS]); first-rotation closure via Stage-Torque Duality (ZS-F15R [35/35 PASS]); the five-phase cyclic cosmology now closes as a fully-derived Möbius ring.

Chapter 9 — Inflation, Reheating, Λ CDM

$n_s = 0.9676$ ($N=60$); $r = 0.00890$; $r_{\text{ZS}} / r_{\text{Staro}} = 2.671 \rightarrow \sim 6\sigma$ LiteBIRD discriminator. $\rightarrow$ ZS-U1 [F-X.1 BLOCKING gate]; ZS-U2 [reheating]; ZS-U4 [global fit, 53/53 PASS].

Chapter 10 — Baryogenesis & QKE

$\eta_B = (6/11)^{35} = 6.117 \times 10^{-10}$. Z_2 symmetry forces both sectors to generate identical asymmetry (PROVEN). Resonant ARS via Pilaftsis-Underwood. $\rightarrow$ ZS-U3, ZS-U7, ZS-S5.

Chapter 11 — Z_2 Mirror Cosmology and Cyclic Phases

$V_E(-\epsilon) \equiv V_E(+\epsilon)$ PROVEN exact. Phases $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow A$ six-step cycle. Sequential X-clock reading (preserved) and parallel dual-clock reading (DERIVED-CONDITIONAL strong via ZS-F10 Theorem F10.2). The 10^{17} ratio $\tau_6/\tau_5 = \exp(\pi/A)$ is the X-observation of the same Y-completion event. $\rightarrow$ ZS-U8 v1.0 [60/60 PASS]; ZS-A8R; ZS-F10 [30/30 PASS]; ZS-F13 [24/24 PASS].

Chapter 12 — Global Fit and Late-Time Cosmology

Ten predictions: (H_0 , Ω_m , w , S_8 , Y_p , D/H , η_B , n_s , r , D_V) confronted with Planck/DES/DESI/SH0ES. $\Delta N_{\text{eff}} = 2A = 0.16018$ resolves D/H from -1.8σ to -0.05σ . $\Omega_\Lambda/\Omega_m = 2e^A$ factor 2 from ZS-F12R Tetrahedral Dual Orientation Multiplicity (NEW v3.0). $\rightarrow$ ZS-U4 [53/53 PASS], ZS-U6 (Cobaya MCMC), ZS-T1.

PART III Source Paper Map (v3.0 extended)

Source paper	Role	Light pointer
ZS-U1	ε -field inflation, F-X.1 BLOCKING gate	$\rightarrow$ ZS-U1
ZS-U2	Reheating dynamics	$\rightarrow$ ZS-U2
ZS-U3	Baryon asymmetry	$\rightarrow$ ZS-U3
ZS-U4	Global cosmological fit, 53/53 PASS	$\rightarrow$ ZS-U4
ZS-U5	Quantum Gravity Bridge, T_d symmetry, Immirzi	$\rightarrow$ ZS-U5
ZS-U6	Cobaya MCMC + face-counting v2.0 $\Omega_{\text{cdm}} = 32/121$	$\rightarrow$ ZS-U6
ZS-U7	QKE-closed baryogenesis	$\rightarrow$ ZS-U7
ZS-U8	Cyclic Holonomy & Z_2 Vacuum, parallel reading DERIVED-COND strong	$\rightarrow$ ZS-U8 [60/60 PASS]
ZS-U9	Hypercharge Trinity Braiding	$\rightarrow$ ZS-U9
ZS-U10	Pentagon Tetraton & Schwinger Coefficient	$\rightarrow$ ZS-U10
ZS-U11	Bounce Q-Survival Closure: V1 Resolution	$\rightarrow$ ZS-U11 [40/40 PASS]
ZS-A8R (post-v2.0)	Contracting Universe Dynamics; Polyhedral-Tetration Bridges	$\rightarrow$ ZS-A8R [22/22 PASS]
ZS-A9R (post-v2.0)	Banach–Tarski origin of doubling-halving symmetry	$\rightarrow$ ZS-A9R [47/47 PASS, +14 v1.0(R) §11]

PART IV — Quantum Gravity Bridge

§0. Abstract

$Q = 11 \leftrightarrow$ quantum tetrahedron (T_d symmetry). Bekenstein-Hawking $\ln 2$ from Z_2 seam parity. Two paths to Immirzi $\gamma_{\text{LQG}} = \ln 2/(\pi\sqrt{3})$. Z-Telomere RG flow, $\lambda_{\text{vac}} = 2A^2$ IR-stable. Cyclic cosmology $S_{Z_2} = 6\pi/A$. Three Polyhedral-Tetration Bridges. Möbius chronology. Banach-Tarski origin of doubling-halving symmetry. $m_e = \sqrt{(5 - \phi)}$. NEW v3.0 (Math Spine RH-Bridge programme): the V_4 -equivariant ZBSI framework on the BV-BFV cobordism-history fiber (M22–M28); the four Dragons D4a–D4d of the Y-sector RH contribution map (M23, sharpened by M25–M28, integrated by M33); the Three-Wall Quantitative Map (M26 Theorem M26.3); and the spinor-cycle averaged closure of the Stapledon Bridge residual (M32) closing NC-M29.RZ-A^PK at DERIVED-CONDITIONAL.

Chapter 13 — $Q = 11$ Spin-Network Correspondence

$|T_d| = 24$, $|\text{Stab}_T(v)| = 6$, $|I_h/T_d| = 5$. $j = 1/2$ unique among half-integers giving $(Z, X, Y) = (2, 3, 6)$ (ZS-M3 Th 5.1). Explicit non-identity: $Q = 11 \neq \dim(\text{intertwiner}) = 2 \neq \dim(\text{Hilbert}) = 16$. $\rightarrow$ ZS-U5 §2-3, ZS-M3.

Chapter 14 — Z_2 Seam Dynamics

Dual κ census (κ_{signflip} vs $\kappa_{\text{disrupted}}$). Regge-Holonomy: $\delta\phi_{\text{cell}} = A$ via 7-step algebraic chain. $SU(2)$ center pairing $D^{\wedge\{1/2\}}(-I) = -I$. $\rightarrow$ ZS-M3 [27/27 PASS]; ZS-Q4.

Chapter 15 — Micro-Macro Transition

Phase drift $\rightarrow$ tunneling action. Z-Anchor dual mechanism. $\lambda_{\text{vac}} = 2A^2$ IR fixed point. Cyclic cosmology $S_{Z_2} = 6\pi/A$. Bounce Q-survival. Three Polyhedral-Tetration Bridges. Möbius chronology. Banach-Tarski origin. $m_e = \sqrt{5 - \phi}$. ZS-M17 continuum-limit theorems M17.1–M17.7 (Tiling Continuum Convergence, Lieb-Robinson Tightness, Reflection Positivity, Path-Integral Closure of $\Delta\Gamma_{G2}$, Pre-Geometric Metric Emergence, Continuum Limit Universality, Wightman QFT Reconstruction). $\rightarrow$ ZS-U5, U8, U11 (closes V1), A6 §4.5.6 (F-A6.1 CLOSED), A8R, A9R, M12, M14, M17.

Chapter 15A — NEW v3.0 — The Math Spine Riemann-Hypothesis Programme (M22–M33)

Cumulative 47+ exploratory rounds consolidated by ZS-M22 v1.1 (Five-Pillar Arithmetic-Dedekind Scaffold) into a non-RH-claiming structural map of where Z-Spin and Riemann ride together (dong-seung). The programme's logical structure:

- ZS-M22 v1.1 [60/60 PASS] — Five Pillars: I ($n=3$ Eisenstein chain), II ($Q=11$ cyclotomic chain), III (multiplicative arithmetic fiber), IV (critical line as J-symmetry locus), V (scalar-kernel no-go for Weil positivity). ADS-6 PROVEN no-go on boundary fiber forces ADS-H1 cobordism BRST positivity hypothesis as the sole surviving route.
- ZS-M23 v1.0(Revised) [31/31 PASS] — Y-Sector RH Contribution Map. Four Dragons D4a (V_4 -Sonin embedding), D4b (Burnol conductor), D4c (defect-square realization), D4d (cobordism BRST-Hodge projection). NC-M23.7: D4 closure alone does not close GRH-for-K.
- ZS-M24 v1.0 [35/35 PASS] — Face polygon spectral zeta $\zeta^*\nabla^*(s)$. Theorem D.1 Legendre duplication archimedean decomposition. Theorem D.2 $B(s)$ identification. P2 status OPEN $\rightarrow$ PARTIAL.
- ZS-M25 v1.0 [26/26 PASS] — Composite biquadratic $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$. Theorem D.1-K four-factor archimedean decomposition $\xi_K(s) = (1/4\sqrt{33}) \cdot \xi(s) \cdot \Lambda(s, \chi_{-3}) \cdot \Lambda(s, \chi_{-11}) \cdot \Lambda(s, \chi_{33})$. Theorem P3-J J-Yakaboylu compatibility (J commutes with $H_Q^{\wedge} \text{Yak}$, J on $\sigma = 1/2$).
- ZS-M26 v1.0 [24/24 PASS] — V_4 -Equivariant ZBSI on cobordism-history fiber. Theorem M26.1 V_4 -Schur idempotent decomposition; Theorem M26.2 Projected Determinant per Channel ($C_\chi \in \{1, \sqrt{3}, \sqrt{11}, \sqrt{33}\}$); Theorem M26.3 Three-Wall Quantitative Map ($W1, W2, W3$).
- ZS-M27 v1.0 [24/24 PASS] — V_4 -Equivariant Cobordism BRST Closure via Kostant Cubic Dirac. Closes Wall $W3$ at DERIVED-CONDITIONAL via three external imports (Alekseev–Barmaz–Mnev 2018, Huang–Pandžić 2002, Kostant 1999/2003).

- ZS-M28 v1.0 [30/30 PASS] — The Z-Spin RH Bridge. Theorem 28.5 $\sigma = 1/2 \leftrightarrow j = 1/2$ isomorphism (DERIVED-interpretation). Corollary 28.5a Möbius-trace infinity (HYPOTHESIS-strong). Theorem 28.14 Y-sector geometric carrier of χ_{-3} (DERIVED-CANDIDATE).
- ZS-M29 v1.0 [79/79 PASS] — Z-Funnel Spectral Retraction with Stapledon Bridge. Theorem 9.1 conditional bridge from string compactification (A_5 -Hilb(X_F) Calabi–Yau threefold, $h^{\{1,1\}} = 5$, $h^{\{2,1\}} = 15$) to the Z-Spin truncated-icosahedron Hodge–Dirac sector. NC-M29.RZ principal OPEN gate.
- ZS-M30 v1.0 [32/32 PASS] — Z-Spin Duality and the RH Bridge. Theorem 30.1 Möbius-Trace Infinity (DERIVED-interpretation strong, six-route over-determination). Theorem 30.2 Y-Sector Two-Face Spectral Mirroring. Theorem 30.5 Twenty-One $1/2$ Catalogue across six categories. Three-layer duality: Sector / Frame / Protocol.
- ZS-M31 v3.0 [36/36 PASS] — W2 as Cross-Coupled Sector Duality. Lemma M31.0 Non-Separability (no sum-form $W_K = F_X + F_Y + F_Z$ compatible with corpus data, max variance 13.011). Theorem M31.4 Z_2 -Parity Selection Rule. Self-Dual Replication Principle (SDRP) registered.
- ZS-M32 v1.0 [56/56 PASS] — Spinor-Cycle Averaged Residual Criterion. Path-Reversal Lemma M32.X ($B^{\wedge}PK$ preserves phase sign, contrasting $B^{\wedge}\dagger$ which flips it). Two-layer phase quantum $\alpha_{amp} = \pi/10$ (amplitude) / $\alpha_{op} = \pi/5$ (operator). $N = 20$ spinor-identity closure: $R_Z^{\wedge}ZS_{\{Z\text{-even}\}^{\{4\pi\text{-avg}\}}} = 0$ EXACTLY at $N = 20$. Closes NC-M29.RZ- $A^{\wedge}PK$ at DERIVED-CONDITIONAL.
- ZS-M33 v1.0 [52/52 PASS] — V_4 -Equivariant Weil Functional Closure via Path γ -revised Z-Mediator Reading. Reading C (NEW): all four V_4 -arithmetic mechanisms route through Z-mediator projector $\Pi_Z = (1/2)(I + J_Z)$. Sub-targets D4a (CCM 2024), D4b (Burnol 1998 + log additivity $\log(3)+\log(11)=\log(33)$), D4c (Wilson-LOCATOR phase factor), D4d (Kostant cubic Dirac BRST cohomology) integrated into single Tr identity. Sign-flip mechanism decomposition ($5/12$ NEG $\rightarrow$ predicted $12/12$ POS).
Programme NON-CLAIM (NC-M23.1 preserved verbatim throughout): Z-Spin does not, and structurally cannot, prove RH from internal data alone. The walls W1, W2, W3 are mapped, sharpened, and progressively closed; the program's contribution is a coordinated three-axis over-determination of the Bridge thesis, not a proof.

PART IV Source Paper Map (v3.0 extended)

Source paper	Role	Light pointer
ZS-U5	Quantum Gravity Bridge, T_d symmetry, Immirzi	$\rightarrow$ ZS-U5
ZS-U8	Cyclic Parallel Reading	$\rightarrow$ ZS-U8 [60/60 PASS]
ZS-U11	Capstone: V1 Planck-bulk CLOSED	$\rightarrow$ ZS-U11 [40/40 PASS]
ZS-M3	Regge-Holonomy + Immirzi + Z-Telomere	$\rightarrow$ ZS-M3 [27/27 PASS]
ZS-M12	Auto-Surgery, $\epsilon_{min} \approx 30.7$	$\rightarrow$ ZS-M12
ZS-M14	Geometric $m_e = \sqrt{(5-\phi)}$	$\rightarrow$ ZS-M14 [47/47 PASS]
ZS-M17	Continuum limit, OS reconstruction	$\rightarrow$ ZS-M17 [60/60 PASS]

ZS-A6	Z-Anchor + cigar bounce, F-A6.1 CLOSED	→ ZS-A6 §4.5.6 [140/140]
ZS-A8R	Polyhedral-Tetration Bridges	→ ZS-A8R [22/22 PASS]
ZS-A9R	Banach-Tarski origin	→ ZS-A9R [47/47 PASS]
ZS-F13	Möbius chronology, ring closure	→ ZS-F13 [24/24 PASS]
ZS-M18 (post-v2.0)	Free-Exploration Log: Speculative Prime-Polyhedral Correspondences	→ ZS-M18 [51/51 PASS]
ZS-M22 v1.1 (post-v2.0)	Five-Pillar Arithmetic-Dedekind Scaffold	→ ZS-M22 [60/60 PASS]
ZS-M23 v1.0(R)	Y-Sector RH Contribution Map; four Dragons D4a–D4d	→ ZS-M23 [31/31 PASS]
ZS-M24	Face polygon spectral zeta + Archimedean Completion	→ ZS-M24 [35/35 PASS]
ZS-M25	Composite-Field $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11}) + J$ -Yakaboylu Bridge	→ ZS-M25 [26/26 PASS]
ZS-M26	V_4 -Equivariant ZBSI on Cobordism-History Fiber	→ ZS-M26 [24/24 PASS]
ZS-M27	Cobordism BRST Closure via Kostant Cubic Dirac (W3)	→ ZS-M27 [24/24 PASS]
ZS-M28	The Z-Spin RH Bridge	→ ZS-M28 [30/30 PASS]
ZS-M29	Z-Funnel Spectral Retraction + Stapledon Bridge	→ ZS-M29 [79/79 PASS]
ZS-M30	Z-Spin Duality and the RH Bridge — 21 halves	→ ZS-M30 [32/32 PASS]
ZS-M31 v3.0	W2 as Cross-Coupled Sector Duality	→ ZS-M31 [36/36 PASS]
ZS-M32	Spinor-Cycle Averaged Feshbach Residual; closes NC-M29.RZ-A^PK	→ ZS-M32 [56/56 PASS]
ZS-M33	Path γ -revised V_4 -Equivariant Weil Functional Closure	→ ZS-M33 [52/52 PASS]

PART V — Astrophysics & Strong-Field

§0. Abstract

Galaxy rotation without dark matter (ε -Halo, $r_Z = 18.3$ kpc). Neutron stars +3.93% mass, tidal $\tilde{\Lambda} +21.24\%$. Black holes: Wald entropy, Z-instanton, sector duality. Information paradox via seam witness; signed witness $\tilde{u}_{\text{seam}}$ (F-A7.3 = cheapest near-term gate). 991/991 PASS cumulative across PARTS V+VI+VII preserved. NEW v3.0: ZS-A8 v1.0(Revised) [22/22 PASS] — contracting-universe dynamics with Polyhedral-Tetration Bridges 1–3 and Symmetry–Asymmetry Unified View; ZS-A9 v1.0(Revised) [47/47 PASS] — Banach–Tarski origin of cosmological doubling-halving symmetry through the $F_2 \rightarrow D_4$ amenability functor and the Julia set $J(T)$ of $T(z) = i^{\wedge}z$ as a measure-theoretic boundary.

Chapter 16 — Galaxy Rotation Without Dark Matter

ε -Halo from Goldstone mode, terminus $r_Z = 18.3$ kpc. Modified gravity $\mu = \Sigma = 1/(1+A)$. Three convergent routes $\Omega_m = 38/121$. Factor 2 in $\Omega_{\Lambda}/\Omega_m = 2e^A$ from ZS-F12R (NEW v3.0: $\mu_{\text{Tet}} = 2$ from Tetrahedral Dual Orientation Multiplicity, F-Tet-2eA-1 REJECTED with

replacement F-Tet-2eA-2). → ZS-A1 [78/78 PASS]; ZS-A5 [50/50 PASS]; ZS-S3 [25/25 PASS]; ZS-F12R [20/20 PASS].

Chapter 17 — Neutron Stars, GW, LSS

NS $M_{\text{max}} + 3.93\%$ uniform across EOS (universal $G_{\text{eff}} = G/(1+A)$). Tidal $\tilde{\Lambda} + 21.24\%$. → ZS-A2.

Chapter 18 — Black Holes (incl. §18.10 Horizon Spinor)

ε -field horizon, Wald entropy, Z-instanton + GW signatures. $\varepsilon(r_H) = 0$ PROVEN (cigar bounce). §18.10 v2.0 mirror of ZS-A7 v1.0(Revised) §3 Theorem 3.2-bis [PROVEN, CPTP/Choi]. → ZS-A3 [49/49]; ZS-A6 [140/140]; ZS-A7R [38/38].

Chapter 19 — Information Paradox Resolved

Seam witness u_{seam} , signed witness $\tilde{u}_{\text{seam}}$, $Q = 11$ register, lattice gauge P1. F-A7.3 = 2π vs 4π discrimination on hardware ~2028-2032 (cheapest gate, ~8% incremental shot cost). → ZS-A4 [54/54]; ZS-A7R §C [38/38].

Chapter 20A — NEW v3.0 — Contracting Universe Dynamics (ZS-A8R)

ZS-A8 v1.0(Revised) [22/22 PASS] introduces the SYMMETRIC contraction dynamics: the wave-to-particle transition controlled by the dual factor $(1-2A)$. Three Bridge formulas: Bridge 1 [OBSERVATION-strong] $|z^*| \approx B = \delta_X + \delta_Y$ at 0.0087%; Bridge 2 [OBSERVATION-strong] $\eta_{\text{topo}} \approx B^2 + (\delta_Y - \delta_X)^2/[Y^2 \cdot (1-2A)]$ at 0.000001% ($Y^2 = 36 = E(\text{TO})$ wave-channel scale); Bridge 3 [DERIVED-CONDITIONAL via slog-L2 equivalence] $\Delta = 1/2 - x^*$ via L2 self-locking iteration at 0.000002%. Y-Time Dilation Theorem 5.3.1 (DERIVED-CONDITIONAL strong via ZS-F10): X-clock observation of Y-completion is dilated by $\exp(\pi/A) \approx 10^{17}$ per added Y-dimension. $\exp(\pi/A) = \exp(N_{\text{CP}} \times \langle \sin^2(\varphi/2) \rangle) = \exp((2\pi/A) \times (1/2))$. Symmetry-Asymmetry Unified View (§SA): the universe tilts simultaneously by $(1+A)$ (X-Inward expansion) and $(1-2A)$ (Y-Outward contraction); only the impedance $A = 35/437$ is frame-invariant.

Chapter 20B — NEW v3.0 — Banach–Tarski Origin of Cosmological Doubling-Halving Symmetry (ZS-A9R)

ZS-A9 v1.0(Revised) [47/47 PASS, 12 OPEN closures] establishes that the $(1+A) \leftrightarrow (1-2A)$ duality of late-time cosmology is the measure-preserving quantization of the Banach–Tarski paradoxical decomposition through the i-tetration fixed point $z^* = -W_0(-i\pi/2)/(i\pi/2)$. Three theorems: ZS-A9.1 (DERIVED) $F_2 \rightarrow D_4$ amenability functor — the BT engine $F_2 \subset \text{SO}(3)$ maps to the amenable register dihedral $D_4 = \langle J, J_Z \rangle$, converting BT non-amenability to Z-Spin amenability; ZS-A9.2 (HYPOTHESIS-strong) the Julia set $J(T)$ of $T(z) = i^z z$ is a measure-zero, totally disconnected, cardinality- $2^{\aleph_0}$ set serving as a constructive ZF-analogue of the BT non-measurable set; ZS-A9.3 (DERIVED-CONDITIONAL) two-branch decomposition $(1+A)(1-2A) = 1 - A(1+2A)$ with X-Inward macroscopic expansion $\exp(A)$, Y-Outward microscopic contraction $Y^2(1-2A)$, and Z-mediator absorbing the residual $A(1+2A) \approx 9.29\%$ via

$\kappa^2 = A/Q = 35/4807$. v1.0(Revised) §11 closes all four OPEN items via the *-homomorphism reformulation, the Świerczkowski geometric realization of free F_2 -rotations, and the inheritance of $dx_X + dx_Y + dx_Z = 0$ from ZS-Q7 §5 master-equation probability conservation. OPEN-2.B (formal ZF/ZFC+AC isomorphism between BT non-measurable and $J(T)$) recategorized as PERMANENT NC under Solovay 1970.

PART V Source Paper Interface (v3.0 extended)

Source paper	Verification	Primary chapter	Key result
ZS-A1	78/78 PASS	Ch 16	Rotation curves, BTFR, M- σ , Keplerian
ZS-A2	verified	Ch 17	NS $M_{\max}$, $\tilde{\Lambda}$, GW $c_T = c$
ZS-A3	49/49 PASS	Ch 18 §1-7	ε -field, Wald, Z-instanton
ZS-A4	54/54 PASS	Ch 19	Seam witness, lattice P1
ZS-A5	50/50 PASS	Ch 16 §16.7	Three convergent $\Omega_m = 38/121$
ZS-A6	140/140 PASS	Ch 18 §18.2 + §18.8	Cigar bounce, F-A6.1 CLOSED
ZS-A7R	38/38 PASS	Ch 18 §18.10 + Ch 19 §19.7-8	Horizon Spinor 4π
ZS-A8R (post-v2.0)	22/22 PASS	Ch 20A	Polyhedral-Tetration Bridges 1–3, Symmetry-Asymmetry View
ZS-A9R (post-v2.0)	47/47 PASS	Ch 20B	Banach-Tarski origin of doubling-halving symmetry
ZS-S3	25/25 PASS	Ch 16 §16.8 + Ch 17 §17.3	Modified gravity
ZS-F12R	20/20 PASS	Ch 16 §16.3	Layer Separation factor 2 origin (μ_Tet)

PART VI — Quantum Mechanics

§0. Abstract

Densest PART [376/376 PASS]. Geometric decoherence $\tau_D/\tau_Penrose = 1/A = 12.49$. Bell CHSH = $2\sqrt{2}$ from $\dim(Z) = 2$. Proton spin Hodge $\frac{1}{2}\Delta\Sigma = 3/22$, $\Delta G = 2/11$. Lattice gauge $a_2 = 19/6$ (TO 0.08 σ Wilson MC). CP Jarlskog from $\delta\phi = A$. Z-Spin Quantum Architecture (QH/QS/QC). Inverse Riemann Engine. Structural Arrow of Time (ZS-Q7). Cross-link with v3.0 RH-Bridge programme (M22–M33) at the ZBSI cobordism-history fiber level.

Chapter 20 — Geometric Decoherence + Born Rule

Stinespring + Lindblad $\Gamma = 2A(\Delta E/\hbar)^2$. $\tau_D = 12.49$. Born rule DERIVED from $L_XY \equiv 0$. NC-F11.1–6 inherited verbatim §20.7. $\rightarrow$ ZS-Q1 [29/29 PASS].

Chapter 21 — Bell, Entanglement, Holographic Conjecture

CHSH = $2\sqrt{2}$, $\tau_ent = \tau/2$. Area Law $\chi = 2$, $\varepsilon_BM \rightarrow 0$. $\rightarrow$ ZS-Q2 [48/48]; ZS-Q6 [42/42].

Chapter 22 — Proton Spin from Polyhedral Hodge

BCC T³ Hodge spectrum, ½ΔΣ = 3/22, ΔG = 2/11. (V+F)_X = 38 Mode-Count Collapse. → ZS-Q3 [40/40].

Chapter 23 — Lattice Gauge QSim on TO

a₂ = (V+F)/G = 19/6 PROVEN topological. Wilson MC TO 0.08σ. J-parity. → ZS-Q4.

Chapter 24 — CP Violation and UV Limits

Jarlskog from δφ = A. UV sufficiency of polyhedral spectra. → ZS-Q5.

Chapter 25 — Bond Dimension, Holographic Area Law, Markov Limit

Bond dimension χ = 2 = dim(Z); Regge area law derived. ε_{BM} = 2/Q Markov limit purely geometric. → ZS-Q6 [42/42 PASS].

Chapter 26 — Structural Arrow of Time

Γ(X→Y)/Γ(Y→X) = dim(Y)/dim(X) = 2 PROVEN; ΔS = ln 2 per Z-transit. Coarse-grained entropy bias from L_{XY} = 0 + dim(X) ≠ dim(Y); microscopic dynamics remain time-reversal symmetric. → ZS-Q7 [40/40 PASS].

PART VI Source Paper Interface

Source paper	Verification	Primary chapter	Key result
ZS-Q1	29/29 PASS	Ch 20	Geometric decoherence, Born rule
ZS-Q2	48/48 PASS	Ch 21	Bell, entanglement, holographic conj.
ZS-Q3	40/40 PASS	Ch 22	Proton spin Hodge
ZS-Q4	verified	Ch 23	Lattice gauge TO
ZS-Q5	verified	Ch 24	CP violation, UV limits
ZS-Q6	42/42 PASS	Ch 25	Area law, Markov limit
ZS-Q7	40/40 PASS	Ch 26	Structural arrow of time
ZS-QH	42/42 PASS	Ch 25/29	Quantum hardware FMD specifications
ZS-QS v1.0(R)	35/35 PASS	Ch 24/29	Inverse Riemann Engine: Triple Structure (DETECTOR + LOCATOR + EXCLUDER) and Boolean Resonance Filter
ZS-QC	52/52 PASS	Ch 29	Quantum architecture system integration

PART VII — Translational and Computational

§0. Abstract

ΔN_{eff} = dim(Z)·A = 2A = 0.16018 → BBN D/H −1.8σ → −0.05σ (ZS-T1). Anti-numerology MC: 5670 × 29 = 164,430 candidates, 237 hits = 7.2% of random expectation 3289

(look-elsewhere PASS — framework MORE constrained than random). Tier-1 ★★★★★ T1-1: $1-n_s = A \cdot x^*$ (0.008%, 0.00σ vs Planck). T1-2: $\alpha_{EM} \approx A/Q$ (0.22% LO, 0.00007% NLO Schur) (ZS-T2). 8/8 closure parameters DERIVED via three chains; $n_s(N^* = 55) = 0.9676$; $r(N^* = 60) = 0.0091$ (ZS-T3). NEW v3.0: ZS-T4 v1.0 [36/36 PASS] — Cosmos-Human Isomorphism: six-step telomere hierarchy and (Body, DNA, Brain) sector decomposition as a two-scale realization of the Z-Spin block architecture; structural, substrate-agnostic, NON-CLAIM that humans physically realize Z-Spin geometry.

Chapter 27 — Phenomenological

$\Delta N_{eff} = \dim(Z) \cdot A = 2A = 0.16018 \rightarrow$ BBN D/H $-1.8\sigma \rightarrow -0.05\sigma$. Block Fiedler Mediation Theorem PROVEN (T1-2/T1-3 reciprocal duality $\lambda_2 = 2A/Q$ with X-face propagator scale $1/\kappa^2 = Q/A \approx 137$ and Y-face vertex coupling $\kappa^2 = A/Q \approx 0.0073$). $\rightarrow$ ZS-T1 [42/42].

Chapter 28 — Spectral Observatory

Anti-numerology MC: 164,430 candidates, 237 hits = 7.2% of random expectation 3289 (look-elsewhere PASS). Tier-1 ★★★★★ T1-1: $1-n_s = A \cdot x^*$. T1-2: $\alpha_{EM} \approx A/Q$ (0.00007% NLO via McKay $c_4 = 4/13$). $\rightarrow$ ZS-T2 [30/30].

Chapter 29 — Z-Sim Forward Simulator + Quantum Architecture

8/8 closure parameters DERIVED via three chains: ZS-Q7 §5.8 + ZS-M3 §10 + ZS-Q5 §5.2. Eigenvalue $\lambda(\lambda + 2A/Q)(\lambda + A) = 0$ to 10^{-17} . 500K MC universal-theorem confirmation. $n_s(N^* = 55) = 0.9676$ PASS Planck; $r(N^* = 60) = 0.0091$ PASS BK18. ZS-QH v1.0 functional material specifications (X = bilayer graphene QD, Z = Bi_2Te_3 topological insulator, Y = C_{60} /CNT phonon reservoir as primary candidate stack); ZS-QS v1.0(Revised) Triple Structure Theorem (DETECTOR confirmed, LOCATOR via argmax, EXCLUDER via argmin) and Boolean Resonance Filter (XOR identity with ZS-F8 (E, R) handshake operators); ZS-QC v1.0 system integration. $\rightarrow$ ZS-T3 [35/35]; ZS-QH; ZS-QS v1.0(R); ZS-QC [52/52].

Chapter 30 — NEW v3.0 — The Cosmos-Human Isomorphism (ZS-T4)

ZS-T4 v1.0 [36/36 PASS] establishes a structural isomorphism between Z-Spin cosmology and the human organism at two parallel levels, with Cardinal NC-4 (Z-Brain) explicitly extended to body and genome — NO physical realization claim. Level I: cosmological timescale hierarchy $\tau_n = t_P \times \exp(n\pi/A)$, $n \in \{0, \dots, 6\}$, mapped onto a six-step human life-cycle (gametic union $\rightarrow$ cleavage $\rightarrow$ stem-cell expansion $\rightarrow$ somatic growth $\rightarrow$ reproductive plateau $\rightarrow$ senescence $\rightarrow$ death/inheritance). Strong anchors at $n=2$ (mediator coset $Z=2 \leftrightarrow$ stem cells, HYPOTHESIS-strong), $n=5$ (proton decay $\leftrightarrow$ senescence, HYPOTHESIS-strong), $n=6$ (Z_2 holonomy $\leftrightarrow$ death/inheritance, HYPOTHESIS-strong). Level II: (X, Z, Y) = (3, 2, 6) sector decomposition mapped onto (Body, DNA, Brain) tripartite anatomy at organism scale. Inherits ZB-P7 v1.0 cosmic-biological VERIFIED status ($\Delta\text{PCI}(\text{Wake}-\text{Anesthesia}) \approx \eta_{\text{topo}}$ at 0.7% empirical / 1.5% in-silico). Hayflick limit $\approx$ Z-Telomere completion count $N_{(2\pi)} = 2\pi/A \approx 78.45$

registered as HYPOTHESIS / TESTABLE-LONG with 5-year derivation-or-retraction horizon.
Twelve falsification gates F-T4.1–F-T4.12 pre-registered.

PART VII Source Paper Interface (v3.0 extended)

Source paper	Theme/Role	Chapter	Verification
ZS-T1	Phenomenological	Ch 27	42/42 PASS
ZS-T2	Observational	Ch 28	30/30 PASS
ZS-T3	Computational (Z-Sim)	Ch 29	35/35 PASS
ZS-T4 (post-v2.0)	Cosmos-Human Isomorphism	Ch 30	36/36 PASS
ZS-F1, F2, F5	Foundations cross-link	all	PROVEN
ZS-M1, M3, M6, M8, M11	Math Spine cross-link	various §	verified
ZS-Q1, Q5, Q7	QM cross-link (chains)	Ch 29 §29.1-4	verified
ZS-U1, U6	Early Universe cross-link	Ch 28 §28.3, Ch 29	verified
ZS-QH/QS/QC	Hardware/Software/Integration translational	Ch 29	42 + 35 + 52 PASS

§VII.B-LIGHT — Seven Critical Near-Term ★ Gates (Timeline-Anchored)

These seven gates will either PASS or FALSIFY Z-Spin within ~10 years. Source: §V.G.6, §VI.G.11, §VII.B. Preserved verbatim from v2.0 Light.

#	Gate ID	Test	Timeline	Status
1	F-A1.14 ★	Gaia DR4 MW outer rotation: Keplerian decline beyond 30 kpc (ϵ -Halo finite vs particle NFW)	~2026	Earliest decisive
2	F-A1.15 ★	Gaia DR4/DR5: r_Z (MW) outside [10, 50] kpc (Z-Spin = 18.3 kpc)	~2026-2030	Pending
3	F-A2.5 ★	DESI/Euclid/Rubin: S_8 outside [0.75, 0.84] (~1% precision)	~2026-2030	Pending
4	F-A7.3 ★	Z-Spin hardware: 2π vs 4π discrimination on signed seam witness $\tilde{u}_{\text{seam}}$	~2028-2032	Cheapest (~8% shot overhead)
5	F-X.1 (LiteBIRD) ★	r outside Z-Spin band 0.0085-0.0094 (BLOCKING; $\sim 9\sigma$ vs Boyle-Finn-Turok $r=0$)	~JFY2032	BLOCKING
6	F-A3.7 ★	Hyper-K: τ_p outside $[10^{33.5}, 10^{35.5}]$ yr (Z-Spin = 2.56×10^{34})	~2030-2035	Pending
7	F-A2.1, F-A2.2 ★	10+ pulsars: NS M_{max} +3.93% uniform; ET/CE: $\tilde{\Lambda}$ +21.24% to ~5%	~2030s, ~2035	Pending

PART VIII — Methodological Toolkit: Solution Tools for Stuck Equations

Purpose. This Part is the operational core of v3.1 Light. It promotes the v3.0 Light from a structural pointer-index ("where to read") to a working solution kit ("how to proceed when stuck"). Each tool below is distilled from a corpus paper that achieved a methodological breakthrough — typically by reframing rather than by adding new mathematics. When an AI agent or human researcher encounters a stuck equation, an OPEN flag, or an apparent contradiction inside the Z-Spin framework, this Part is the first place to consult.

Reading order. PART VIII is organised as nine numbered tools (T1–T9). Each tool is given in a fixed schema: (a) Trigger — the symptom that suggests this tool; (b) Source — the originating paper and §-anchor; (c) Statement — the operational kernel in compressed form; (d) Worked corpus precedent — the OPEN flag the tool was first used to close; (e) Generalisation — the broader class of stuck equations the tool can be re-applied to. PART IX (Insight Gallery) extends the tool kit with the deeper conceptual reframings (sector-equivalences, frame-transformations, cross-coupling), and the Master Bypass Index (after PART IX) provides the symptom → tool lookup table.

§VIII.1 — Tool T1: Five-Axiom Diagnostic Lattice (F18)

Trigger.

(a) An equation appears stuck and the failure mode is unclear; (b) a derivation looks circular; (c) several papers address "the same question" but their results don't compose. In any of these cases, T1 isolates which of the five structural axioms is failing.

Source.

ZS-F18 v2.1 §3 + §6.6 (Table 6.1) + §7.4. The five axioms $\mathcal{A} = \{\text{Existence, Self-reference, Algebra, Non-triviality, Unitarity}\}$ and the Four Bridges from $\mathcal{A}$ to the operational corpus.

Statement (operational form).

When stuck, ask which axiom is the failure attributable to. Each axiom has a Bridge (theorem) into the corpus; locate the Bridge and ask whether the bridge is intact for the present problem. The five-axiom set is exhaustive for Z-Spin physical content (ZS-F18 §5, INTERPRETATION-strong); a stuck equation that does not map to any of the five is either a numerology candidate (Anti-Numerology Gate, ZS-F2) or a category error (frame mismatch — see Tool T6).

Axiom	Failure symptom	Controlling Bridge	First port of call
A1 Existence	Object referenced but not constructed	FFPP nested compression	ZS-F0 §13 Theorem 13.3
A2 Self-reference	Definition uses what it defines (no fixed-point)	Bridge One (Cross-Coupling)	ZS-M2 §5 PROVEN; ZS-M1 i-tetration
A3 Algebra	Numerical balance fails / numerology	Master Equation (algebraic)	ZS-M1 §4 + ZS-F17 §3
A4 Non-triviality (algebraic)	Two-channel interface collapsed	Bridge Two (NOT-AND handshake)	ZS-F8 §4 PROVEN
A4 Non-triviality (geometric)	Dimension $\dim(X)=3$ not derived	Bridge Four (Commutator Generator)	ZS-F18 §6 + ZS-S15 §3
A5 Unitarity	Probability/information not preserved	Bridge Three (Z-Bottleneck)	ZS-Q7 Thm 2 + ZS-F8 §4

Sixth polarity	Discrete vs continuous mismatch	X-Y Tiling Asymmetry	ZS-M6 §5.5 PROVEN; ZS-M17
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Worked corpus precedent.

F18 itself diagnosed 12 of the corpus's most consequential identifications (ZS-F18 Appendix A) by mapping each to its responsible axiom. The diagnosis produced no new theorem yet revealed why each result is robust — a model usage of T1 as a methodological tool.

Generalisation.

Any future Z-Spin paper encountering an OPEN flag should run T1 first. If the axiom is identified, the closure path is the corresponding Bridge; if no axiom matches, the OPEN is either out-of-scope (NON-CLAIM territory, Tool T7) or a category error (Tool T6, frame mismatch).

§VIII.2 — Tool T2: Dimensional Reduction via Bridge Four (F18)

Trigger.

Stuck on a high-dimensional construction (dim 3, 6, or 11) where the dependency chain becomes intractable. T2 reduces the problem to $\dim(Z) = 2$ — the only foundational dimension — and lets $\dim(X)=3$ and $\dim(Y)=6$ re-emerge by Lie commutator and product respectively.

Source.

ZS-F18 §6.4–§6.7 (Bridge Four, dimensional origin table). ZS-S15 §3 Theorem S15.2 (Poynting–Commutator Theorem, DERIVED).

Statement.

Two J-conjugate 2π SO(3) circulations of the twin-Reuleaux pair (V_{XZ} , V_{ZY}), composed at 90° NOT-AND phase, generate a third axis $J_S = [J_{R_1}, J_{R_2}]$ by Lie commutator. Therefore $(Z, X, Y) = (2, 3, 6)$ is generated from $\dim(Z) = 2$ alone; $\dim(Y) = \dim(X) \cdot \dim(Z) = 6$; $Q = Z + X + Y = 11$.

Worked corpus precedent.

ZS-F18 v2.0 used T2 to read the question "why does physical space have dimension three?" not as an additional axiom but as a derived consequence of $\dim(Z) = 2$ plus 90° NOT-AND composition. This converted what would have been a 13th independent encounter into the geometric face of axiom A4.

Generalisation.

When facing a stuck calculation in 3D or 6D, examine whether the structure can be re-expressed as two $\dim(Z) = 2$ channels meeting at orthogonal phase. If yes, the third axis is automatic. Application targets include any future SO(3) or SU(2)-related symmetry breaking in PARTS II–V where dimensional emergence is at stake.

§VIII.3 — Tool T3: Hermitian-to-Path-Reversal Switch (M32)

Trigger.

A Feshbach-type residual or sandwich expression refuses to vanish under the standard Hilbert-space adjoint reading. The phase factors cancel "too cleanly" or "not at all", giving either a trivial-positive or unbounded-positive result that doesn't engage with the corpus structure.

Source.

ZS-M32 v1.0 §2 (R_Z^H vs R_Z^{ZS} separation), §3 (Path-Reversal Lemma M32.X). Underpinned by ZS-S6 §4.1 $K_{bwd} \neq K_{fwd}^\dagger$ PROVEN at kernel level and ZS-T1 §10.5 PK-Conjugation Theorem T9 DERIVED.

Statement (Path-Reversal Lemma M32.X).

Under the corpus PK-Conjugation T9 plus the Regge T-odd scalar mechanism, $B \propto e^{+ik\alpha} \Rightarrow B^{PK} \propto e^{+ik\alpha}$, in contrast with the standard adjoint $(e^{+ik\alpha} \cdot B)^\dagger = e^{-ik\alpha} \cdot B^\dagger$. The path-reversal residual $R_Z^{ZS} = B \cdot Q^{-1} \cdot B^{PK}$ is a structurally distinct object from the Hermitian residual $R_Z^H = B \cdot Q^{-1} \cdot B^\dagger$; averaging-and-vanishing claims hold for R_Z^{ZS} , not R_Z^H .

Worked corpus precedent.

Closed NC-M29.RZ-A^{PK} at DERIVED-CONDITIONAL: $R_Z^{ZS}|_{\{Z\text{-even}\}^{4\pi\text{-avg}}} = 0$ EXACTLY at $N = 20$. The Stapledon Bridge (ZS-M29 Theorem 9.1) was upgraded from DERIVED-CONDITIONAL to DERIVED in the spinor-cycle averaged sense without any external numerical input. The closure was achieved purely by switching the operator-reading from R_Z^H to R_Z^{ZS} .

Generalisation.

Any sandwich expression in the framework that fails to close under Hermitian reading should be re-examined under PK reading. Candidate targets: future Feshbach-like reductions in PARTS IV–VI; any operator residual that involves K_{bwd} (T-odd) on one side and K_{fwd} on the other.

§VIII.4 — Tool T4: Two-Layer Phase Quantum Decomposition (M32)

Trigger.

Two corpus values that should be equal differ by exactly a factor of 2; or, an unexpected factor 2 appears that cannot be attributed to $SO(3)/SU(2)$ double-cover. T4 separates amplitude-layer quanta from operator-layer quanta.

Source.

ZS-M32 §4 (two-layer phase quantum). Independent corpus inputs: ZS-S6 §G.2 PROVEN (vertex Regge deficit $\delta_X^{\text{vertex}} - \delta_Y^{\text{vertex}} = \pi/6 - \pi/15 = \pi/10$) and ZS-S6 §G.4 reading ii PROVEN (Z-quarter-turn Z_5 partition $(1/4)(2\pi/5) = \pi/10$).

Statement.

Z-Spin operator-level phase quantum $\alpha_{op} = \pi/5$ decomposes as $\alpha_{op} = 2 \cdot \alpha_{amp}$ where $\alpha_{amp} = \pi/10$ is the amplitude-level quantum. The factor 2 between layers is forced by the Path-Reversal Lemma applied to the Feshbach sandwich; it is the operator-layer analogue of the $SO(3)/SU(2)$ double-cover (ZS-S15 §5).

Worked corpus precedent.

Reconciled four PROVEN occurrences of $\pi/10$ (vertex Regge, Z-quarter-turn) with PROVEN occurrences of $\pi/5$ (commutator $[\sigma_3, \hat{n}_Y] = -2i \cdot \sin(\pi/5) \cdot \sigma_2$ in ZS-S6 §3.3, BCH leading order in ZS-S6 §3.4) under a single layer-doubling identity. Before T4, these were treated as independent constants.

Generalisation.

When two PROVEN corpus quantities differ by factor 2 with no obvious double-cover origin, check whether one is an amplitude-layer quantum and the other an operator-layer quantum. The factor 2 is then the structural signature of the layer transition, not a free parameter or coincidence.

§VIII.5 — Tool T5: Spinor-Cycle Averaging at N = 20 (M32)

Trigger.

A residual or sum that does not vanish at single iterate, but whose single-iterate phase is a primitive 10th root of unity. The standard algebraic minimum $N = 10$ produces zero algebraically but leaves the spinor in fermion-sign-flipped state $D^{\{1/2\}}(2\pi) = -I$, breaking physical closure.

Source.

ZS-M32 §5 (Theorem M32.3, spinor-cycle averaged vanishing). ZS-M3 Lemma 10.1 PROVEN ($D^{\{1/2\}}(4\pi) = +I$). ZS-S15 §5 ($SO(3)/SU(2)$ double-cover ratio).

Statement.

When $R_{Z^{\text{AS}}}(k) \propto \exp(ik\pi/5) \cdot R_{Z^{\text{AS}}}(0)$, the geometric series $\sum_{k=0}^{19} \exp(ik\pi/5) = 0$ EXACTLY, but only at $N = 20$ (not at $N = 10$) does the cumulative spinor phase reach 4π , restoring $D^{\{1/2\}}(4\pi) = +I$. The minimum physical-closure averaging window is $N = 20$.

Worked corpus precedent.

$R_{Z^{\text{AS}}} |_{\{Z\text{-even}\}^{4\pi\text{-avg}}} = 0$ EXACTLY at $N = 20$. ZS-M32 Table 7 explicitly compares $N \in \{10, 20, 30, 40\}$ and identifies the minimum physical closure as $N = 20$. The averaging structure aligns with three other corpus 4π averages: $\langle \sin^2(\varphi/2) \rangle_{[0,4\pi]} = 1/2$ (ZS-M3), $\langle V_{XZ} \cdot V_{ZY} \rangle = 1$ (ZS-T1), $\langle \varepsilon_+ + \varepsilon_- \rangle = 0$ (ZS-U1) — Table 8 of ZS-M32.

Generalisation.

Any non-vanishing single-iterate quantity whose phase is a primitive 10th root of unity admits closure under $N = 20$ spinor-cycle averaging. Candidate targets: future Feshbach-like reductions in PARTS IV/VI; any operator-residual whose phase quantum is $\alpha_{\text{op}} = \pi/5$.

§VIII.6 — Tool T6: Three-Way Split of OPEN Flags (M32)

Trigger.

An OPEN flag too large to close in one step. Attempting full closure produces circularity or external-import-blocking. T6 splits the OPEN into three structurally distinct sub-claims with different status levels.

Source.

ZS-M32 §6 (NC-M29.RZ three-way split). General pattern adoption from ZS-M22, ZS-M23R Four Dragons D4a–D4d.

Statement.

An OPEN flag NC-X may be split as: A^{PK} = the part closeable under path-reversal/averaging now (DERIVED-CONDITIONAL); B = the part with structural evidence but one identified gap (HYPOTHESIS-strong, with explicit closure path); C = the instantaneous/full-spectrum part requiring external numerical or theoretical input (OPEN, inherited).

Worked corpus precedent.

NC-M29.RZ $\rightarrow \{A^{\text{PK}} = R_{Z^{\text{AS}}} |_{\{Z\text{-even}\}^{4\pi\text{-avg}}} = 0 \text{ at } N = 20 \text{ (DERIVED-COND.)}; B = \text{single-iterate magnitude (HYPOTHESIS-strong, bulk-spectral-gap closure path)}; C = R_{Z^{\text{AS}}} |_{\{\text{full}\}} \text{ on Ricci-flat metric (OPEN, ML-CY criterion)}\}$. Promotes ZS-M29 Theorem 9.1 from DERIVED-CONDITIONAL to DERIVED in averaged sense without closing C .

Generalisation.

Any OPEN flag should be split before closure attempt. The three sub-statuses ($A^{\text{closeable}}$ / $B^{\text{one-gap}}$ / $C^{\text{inherited}}$) prevent the all-or-nothing trap that would otherwise leave an entire programme blocked on a single external computation.

§VIII.7 — Tool T7: Cross-Coupling Discipline (M31)

Trigger.

An expression naturally factors as $W = F_X + F_Y + F_Z$ (sum-form) but the per-channel data refuse to compose. The summands appear independent, yet their sum disagrees with measured cross-channel variance. A factorisation appears, but it is the wrong factorisation.

Source.

ZS-M31 v3.0 §4.0 Lemma M31.0 (Non-Separability). ZS-M2 §5 Cross-Coupling Theorem PROVEN.

Statement (Lemma M31.0, Non-Separability).

There is no decomposition $W_K(g) = F_X + F_Y + F_Z$ compatible with the ZS-M26 PROVEN per-channel data; the maximum across-channel variance is 13.011 across 18 (a, t1, t2) test points. The correct form is the bilinear cross-coupled expression $W_{XYZ}(g) := \langle g_X, (\Pi_Z \otimes I_{\text{arith}})(B_Y - P_Y)(\Pi_Z \otimes I_{\text{arith}}) g_X \rangle$ where $\Pi_Z = (1/2)(I + J_Z)$ is the Z-mediator projector.

Worked corpus precedent.

The W2 wall of the Z-Spin RH program had two competing readings before M31: Reading A (Y-incompleteness) and Reading B (archimedean+conductor superposition). Both are sum-forms; Lemma M31.0 falsifies both as Cross-Coupling Theorem violations and forces a bilinear cross-coupled reading. The correct W2 form is Sector-Duality ($X \otimes Y$ mediated by Z), not a sum.

Generalisation.

Whenever a corpus expression is a candidate for sum-form decomposition, run Cross-Coupling Discipline: check whether the per-channel data composes additively or only via the Z-mediator bilinear. The default assumption in Z-Spin should be the latter; sum-forms are the exception, not the rule.

§VIII.8 — Tool T8: Z-Mediator Projector Routing (M33)

Trigger.

Multiple sub-targets that each individually engage a different external mathematical tradition refuse to integrate into a single closed identity. T8 routes all sub-targets through the Z-mediator projector Π_Z , producing a single trace identity.

Source.

ZS-M33 v1.0 (Reading C). $\Pi_Z = (1/2)(I + J_Z)$ defined in ZS-M31 §4.0 Definition M31.0c. Path-γ-revised reading specific to M33.

Statement.

All four V_4 -arithmetic mechanisms (D4a CCM 2024 Sonin, D4b Burnol conductor, D4c Wilson-LOCATOR, D4d Kostant cubic Dirac) route through the projector $\Pi_Z = (1/2)(I + J_Z)$ into a single Tr identity. The sign-flip mechanism on the W2 wall decomposes 5/12 NEG into the predicted 12/12 POS structure under Reading C.

Worked corpus precedent.

Before M33, D4a–D4d were treated as four parallel sub-targets, each with its own external mathematical engagement. Reading C unifies them into a single bilinear Tr identity, eliminating four-way bookkeeping and replacing it with a single Π_Z -projected trace. The four prefactors $\{1, \sqrt{3}, \sqrt{11}, \sqrt{33}\}$ (PROVEN, ZS-M22 §7.2) appear naturally as the V_4 -decoration of a single underlying identity.

Generalisation.

Whenever multiple sub-targets refuse to integrate, examine whether each can be expressed as a Π_Z -projection of a single underlying Tr expression. The Z-mediator projector is the corpus's universal routing operator: it lives on $\dim(Z) = 2$ and carries the full information cost of cross-sector mediation.

§VIII.9 — Tool T9: Self-Referential Fixed-Point Closure (F11)

Trigger.

An OPEN flag involves an observer or a coordinate frame whose definition appears circular. The naive resolution introduces a phenomenological observer (a frame from outside the corpus); T9 closes the OPEN by promoting the observer to an operationally defined self-referential fixed point internal to the corpus.

Source.

ZS-F11 v1.0 §5 (OOC convergence) + §6 (six explicit NON-CLAIMs against phenomenological consciousness).

Statement.

The Operational Observer Coordinate (OOC) is defined as the self-referential fixed point of the framework's measurement protocol; it is constructed from PROVEN corpus inputs only. The OOC closes ZS-M11 §H16 (formerly OPEN) without introducing any frame from outside the corpus. Six NON-CLAIMs are explicitly registered to prevent overclaim (NC-F11.1–NC-F11.6).

Worked corpus precedent.

ZS-M11 §H16 OPEN flag (phenomenological observer reference) was a long-standing obstacle in the framework. ZS-F11 closed it by replacing the phenomenological observer with an operationally defined fixed point of the measurement-action iteration, rendering the closure entirely internal to the corpus.

Generalisation.

When an OPEN flag tempts the introduction of an external/phenomenological agent, ask first whether the agent can be replaced by a self-referential fixed point internal to the framework. The OOC pattern is the canonical method: define the agent as the convergent limit of an iteration whose generators are corpus-PROVEN.

§VIII.10 — Methodological Toolkit Summary Table

Tool	One-line trigger	Source	Output status
T1 Five-Axiom Diagnostic	Stuck / circular / non-composing	F18 §3+§6	Identifies failing axiom

T2 Bridge Four	High-dim construction intractable	F18 §6 + S15 §3	$\dim(X)=3$ from $\dim(Z)=2$
T3 H→PK Switch	Hermitian sandwich won't close	M32 §2-3	Different operator class
T4 Two-Layer Phase	Unexplained factor 2 between corpus values	M32 §4	$\alpha_{op} = 2 \cdot \alpha_{amp}$ identity
T5 N=20 Averaging	Single iterate non-vanishing, primitive 10th root	M32 §5	0 EXACTLY at N=20
T6 Three-Way Split	OPEN flag too large	M32 §6	$\{A^{PK} / B / C\}$ sub-flags
T7 Cross-Coupling Discipline	Sum-form fails per-channel data	M31 §4.0 Lemma M31.0	Bilinear W_{XYZ} form
T8 Π_Z Routing	Multiple sub-targets won't integrate	M33 Reading C	Single Tr identity
T9 OOC Fixed-Point	Phenomenological observer tempted	F11 §5-6	Self-referential closure

PART IX — Insight Gallery: Re-Interpretation Methodology

Purpose. Where PART VIII collects nine operational tools ("how to do X"), PART IX collects six methodological insights from the corpus's most ambitious re-interpretation papers ("how to think about Y"). These are reading lenses, not calculation tools. Each lens is a way of looking at the corpus that resolves tensions which would otherwise be visible only as paradoxes. They are organised as I1–I6 and reference the source papers from which each lens was distilled.

§IX.1 — Insight I1: Symmetry-Asymmetry Unified View (A8R §SA)

Lens. Every Z-Spin observable is a deviation from the would-be balance of 1 (unity) and $1/2$ (Z_2 -reflection fixed point). The universe's entire dynamics is the composition of sectoral tilts of $A = 35/437$ through the (X, Y, Z) decomposition. $(1+A)$ is X-Inward expansion; $(1-2A)$ is Y-Outward contraction; $(1+A)(1-2A) = 1 - A(1+2A) \approx 0.9071$ measures the joint deviation from balance at $\approx 9.29\%$.

Why it is a tool. A8R §SA replaces the apparent independence of expansion and contraction physics with a single tilted-balance reading. The 10^{17} ratio $\tau_6 / \tau_5 = \exp(\pi/A)$ is then read as the X-frame observation of the same Y-completion event. Future Z-Spin papers should default to this lens for any cosmological / lifecycle / time-scale question.

Status. INTERPRETATION, HYPOTHESIS-strong (A8R §SA.7); 13 explicit corpus connections established (A8R §SA.6).

§IX.2 — Insight I2: Three-Layer Duality (M30 §6)

Lens. Every apparent contradiction in Z-Spin Cosmology is resolved as one object viewed from two frames. The duality operates at three coordinated layers: Sector Duality ($X \leftrightarrow Y$ mediated by Z), Frame Duality (X-observer $\leftrightarrow$ Y-observer; ZS-A8 §SA.4), Protocol Duality (state-preparation $\leftrightarrow$ spectral characterisation; ZS-F16 Two-Protocol Theorem).

Why it is a tool. The three RH walls of ZS-M28 (W1, W2, W3) — formerly read as unstructured "problems" — are reread under this lens as duality interfaces: W1 = Frame Duality, W2 = Sector Duality, W3 = Protocol Duality. The lens does not by itself close the walls; it places each wall in a definite Layer with a definite external-mathematics import path. This converts an unstructured OPEN list into a structured navigation map.

Status. DERIVED-interpretation strong (M30 Theorem 30.5 + §6).

§IX.3 — Insight I3: Six-Route Möbius-Trace Infinity (M30 §3)

Lens. The pattern "external count-infinity $\leftrightarrow$ internal closure" appears in the corpus six independent times (Riemann zero sequence $\leftrightarrow T(T(z^*)) = z^*$; cosmological cycle index $\leftrightarrow A$ frame-invariant; BV-BFV ghost tower $\leftrightarrow 4\pi$ closure; FFPP nested compression $\leftrightarrow W_0(-i\pi/2)$; Z-mediation outcomes $\leftrightarrow \ln(2)$ bit per pass; Wilson cycle ergodic limit $\langle \cos^2(n \cdot 129.45^\circ) \rangle \rightarrow 1/2$). Recognising the six as one pattern raises the Möbius-trace infinity reading from HYPOTHESIS-strong to DERIVED-interpretation strong (M30 Theorem 30.1).

Why it is a tool. The lens identifies that what an external observer counts as "infinitely many things" the framework reads as a single closed loop traversed with a Möbius-type frame transformation. It is the standard reading lens to apply whenever the corpus generates a sequence-cardinality that an external observer might mistake for additional structure beyond the framework.

Status. DERIVED-interpretation strong with six-route over-determination (M30 §3 Table 3.1 + Theorem 30.1).

§IX.4 — Insight I4: Twenty-One $\frac{1}{2}$ Catalogue (M30 §3.5)

Lens. The corpus contains exactly 21 PROVEN/DERIVED instances of the fraction $1/2$, organised in six structural categories. Combined with Three 2s Unity (Taylor-2, Bottleneck-2, Sector-2 all equal $\dim(Z) = 2$; ZS-A8 §SA.3 + Test K1 PASS), the corpus contains $21 + 3 = 24$ instances of $\dim(Z) = 2$ manifesting at different dimensional and contextual levels. All 21 are dimensional projections of the same Z_2 involution $x \leftrightarrow 1 - x$.

Why it is a tool. When a future Z-Spin construction produces a $1/2$, the catalogue is the first reference to consult — to determine whether the new $1/2$ is one of the existing 21, or constitutes a 22nd entry (which would require its own verification gate). The catalogue prevents double-counting of the same Z_2 -fixed-point structure under different names.

Status. DERIVED-interpretation (M30 Theorem 30.5).

§IX.5 — Insight I5: Four-Layer 50/50 Equivalence (F17)

Lens. The four 50/50 partitions in the corpus — (3D) truncated tetrahedron 4+4 split; (2D) Reuleaux $\pi+\pi$ curvature distribution; (1D) Master Equation $|Term A| = |Term B|$; (0D) XOR alphabet E vs R — are dimensional projections of a single per-cycle equilibrium between information accumulation and information release in the Z-mediated bottleneck channel.

Why it is a tool. ZS-F17 Theorem 5.1 (Self-Referential Information-Compression Equilibrium) shows that the 50/50 partition is PROVEN at the equation level ($|Term A| - |Term B| = O(10^{-51})$ at 50-digit precision), not measured. When a future Z-Spin construction produces a 50/50 partition, the lens identifies which of the four dimensional layers it lives at and what it accumulates ($\ln(2)/3$ nats per

vertex matches the Z-bottleneck capacity bound, ZS-Q7 Theorem 2). This promotes the Reuleaux 50/50 from HYPOTHESIS (NC-F7.5) to DERIVED-CONDITIONAL.

Status. DERIVED-CONDITIONAL strong (F17 Theorem 5.1, over-determined by four independent corpus PROVEN inputs).

§IX.6 — Insight I6: Banach-Tarski Origin of Doubling-Halving (A9R)

Lens. The $(1+A) \leftrightarrow (1-2A)$ duality of late-time cosmology is the measure-preserving quantization of the Banach-Tarski paradoxical decomposition through the i-tetration fixed point $z^* = -W_0(-i \pi/2)/(i \pi/2)$. The non-amenable BT engine $F_2 \subset SO(3)$ is mapped via a functor to the amenable register dihedral $D_4 = \langle J, J_Z \rangle$, converting BT non-amenableity to Z-Spin amenability. The Julia set $J(T)$ of $T(z) = i^{\wedge}z$ is the constructive ZF-analogue of the BT non-measurable set (measure-zero, totally disconnected, cardinality $2^{\aleph_0}$).

Why it is a tool. The lens supplies a structural origin (rather than a phenomenological one) for the doubling-halving symmetry that pervades Z-Spin cosmology. When a future paper finds a doubling-halving identity, the lens directs the search for its measure-theoretic boundary in $J(T)$ — the Z-Spin amenable analogue of the BT non-measurable set.

Status. ZS-A9.1 DERIVED, ZS-A9.2 HYPOTHESIS-strong, ZS-A9.3 DERIVED-CONDITIONAL. OPEN-2.B (formal ZF/ZFC+AC isomorphism) recategorised as PERMANENT NON-CLAIM under Solovay 1970.

Master Bypass Index — Symptom → Tool Lookup

When a stuck equation, OPEN flag, or apparent contradiction is encountered, match the symptom in the leftmost column to the recommended tool/insight. The rightmost column gives the canonical worked example — the corpus instance where the tool was first deployed.

Symptom	First tool	Backup lens	Canonical example
Stuck equation, failure mode unclear	T1 (F18 §3,§6)	I2 (M30 §6)	F18 Appendix A — 12 encounters
dim 3, 6, or 11 construction intractable	T2 (F18 §6, S15 §3)	I5 (F17)	S15 §3 $J_S = [J_{R_1}, J_{R_2}]$
Hermitian sandwich phase-cancels trivially	T3 (M32 §2-3)	I2 W2 (M30 §6)	M32 closing NC-M29.RZ-A ^{PK}
Unexpected factor 2 between corpus values	T4 (M32 §4)	T2 (Bridge Four μ_{Tet})	M32 $\alpha_{op} = 2 \cdot \alpha_{amp}$ identity
Single iterate non-vanishing, primitive 10th root phase	T5 (M32 §5)	I3 (M30 §3)	M32 $R_{Z^{\wedge}ZS} ^{\wedge\{4\pi-avg\}}$ at N=20
OPEN flag too large to close in one step	T6 (M32 §6)	I2 (M30 §6 walls)	NC-M29.RZ split A ^{PK} /B/C
Sum-form decomposition fails per-channel	T7 (M31 Lemma M31.0)	I2 (Sector Duality)	M31 W2 cross-coupled bilinear
Multiple sub-targets won't integrate	T8 (M33 Reading C)	I2 (M30 §6 layers)	M33 D4a-D4d → single Tr identity

Phenomenological observer tempted	T9 (F11 §5-6)	I1 (A8R §SA Frame Equiv.)	F11 closing M11 §H16
Apparent independence of expansion / contraction	I1 (A8R §SA)	I6 (A9R)	A8R §SA Symmetry-Asymmetry View
External count-infinity in derivation	I3 (M30 §3)	I2 (M30 §6)	M30 six-route Möbius-trace
A new 1/2 appears, possibly redundant	I4 (M30 §3.5 Catalogue)	I5 (F17)	M30 21·½ Catalogue
A new 50/50 partition appears	I5 (F17 Theorem 5.1)	I4 (M30 §3.5)	F17 four-layer equivalence
Doubling-halving identity needs origin	I6 (A9R)	I1 (A8R §SA)	A9R $F_2 \rightarrow D_4$ functor
Quantitative wall in RH programme	T7 + T8 combined	I2 layered reading	M26 §5.5 + M33 Reading C
Need to falsify candidate 1-loop closure	Anti-Numerology Gate (F2)	T1 (Algebra axiom)	M21 sextic falsifies M20.A

Note. The Master Bypass Index is intentionally a flat lookup. For combined symptoms (e.g., "stuck Feshbach sandwich with factor-2 mismatch and primitive-10th-root phase"), apply the tools in the order $T3 \rightarrow T4 \rightarrow T5$; this is the precise composition that ZS-M32 used to close NC-M29.RZ-A^{PK}.

Master Falsification Registry (v3.1 Summary, preserved from v3.0)

Across 101 papers the Z-Spin Cosmology programme has accumulated a verification base of approximately 1,800+ PASS-rated checks, with no failures recorded against any DERIVED-locked identity. The registry below summarises the most consequential falsifiable entries — those whose negation would invalidate a DERIVED, BRIDGE, or TESTABLE claim of the framework. Each entry points to the originating paper and the canonical formulation; resolution of any single OPEN entry below is sufficient to either consolidate or refute a major branch.

§ Anchor	Falsifiable Statement	Status (v3.0)
§F1.5 [LOCKED]	Z-Field Master Equation $A=p_X \cdot p_Y$	DERIVED — F9 §3, S15 §B
§F2.7 [LOCKED]	$(p-1)(q-1)=p \Rightarrow$ unique solution (2,3)	DERIVED — M19 Forcing Theorem
§F8.6 [LOCKED]	Y-Time Dilation $\exp(\pi/A) \approx 10^{17}$	DERIVED — A8R §SA.7, F10 §4
§F11.3 [LOCKED]	OOB fixed point closes M11 §H16	DERIVED — F11 §5
§F12.4 [LOCKED]	Tetrahedral orientation multiplicity $\mu_{\text{Tet}}=2$	DERIVED — F12 §3, originates $2e^A$ factor
§F14.3 [LOCKED]	Kepler-Lyapunov Conjugate Uniqueness closes F-F7.11	DERIVED — F14 Strategy B
§F15.2 [LOCKED]	$\theta(0) \cdot T_P = A$ unit-fixing identity	DERIVED — F15R §2

§F18.4 [LOCKED]	Bridge Four: Lie commutator generates $\dim(X)=3$ from $\dim(Z)=2$	DERIVED — F18 §4 (Commutator Generator)
§M16.7 [LOCKED]	$\xi_K(s)=(1/4\sqrt{33})\cdot\xi(s)\cdot\wedge(\chi_{-3})\cdot\wedge(\chi_{-11})\cdot\wedge(\chi_{33})$	DERIVED — M25 J-Yakaboylu Bridge
§M17.4 [LOCKED]	$\sigma_1/\sigma_2=Q+Y=17,$ $\sigma_1/\sigma_3=(Q+Y)^2\cdot G+(Q-Z)=3477$	DERIVED — M20 EXACT (Sextic falsifies 1-loop CW closure)
§M27.3 [LOCKED]	V_4 -Equivariant BRST closure of W3 via Kostant Cubic Dirac	DERIVED-CONDITIONAL — M27 §4
§M28.5 [LOCKED]	$\sigma=1/2 \leftrightarrow j=1/2$ RH-Bridge correspondence	DERIVED — M28, M30 (Six-route Möbius-trace)
§M31.0 [LOCKED]	Lemma M31.0 Non-Separability of W2 cross-coupling	DERIVED — M31 §0
§M32.X [LOCKED]	Path-Reversal Lemma: $R_Z^{\wedge ZS} _{\{Z\text{-even}\}}^{\wedge\{4\pi\text{-avg}\}}=0$	DERIVED — M32 EXACT
§M33.C [LOCKED]	Path γ -revised Reading C integrates D4a-D4d	DERIVED — M33 §C, BV-BFV scaffold
§A8R-SA [LOCKED]	Symmetry-Asymmetry Unified View, Three Bridges	DERIVED — A8R 22/22 PASS
§A9R-2 [LOCKED]	$F_2 \rightarrow D_4$ functor, Julia $J(T)$ ZF-analogue of BT non-measurable	DERIVED — A9R §2
§S15.5 [LOCKED]	Maxwell from Twin-Reuleaux handshake; $SO(3)/SU(2)$ factor 2	DERIVED — S15 §5
§T4.6 [TESTABLE-LONG]	Hayflick limit $\approx$ $N_{(2\pi)}=2\pi/A \approx 78.45$ (NC-4 NO physical realization claim)	TESTABLE-LONG — T4 §6
§F-F7.11 [CLOSED]	Twin-Reuleaux Midpoint conjugate decomposition	RESOLVED — closed by F14 v3.0
§M11-H16 [CLOSED]	Phenomenological observer reference	RESOLVED — closed by F11 v3.0
§M22-W3 [PROMOTED]	ADS-H1 cobordism BRST hypothesis	DERIVED-CONDITIONAL — promoted via M27
§M22-W2 [PROMOTED]	Cross-Coupled Sector Duality wall	DERIVED — promoted via M31
§M21.NoGo	1-loop CW closure for M20.A FALSIFIED	FALSIFIED — M21 sextic invariant theory

Note. The single FALSIFIED entry (M21.NoGo) is by design: it falsifies a candidate 1-loop Coleman-Weinberg closure for the M20.A pathway, and its falsification is the very content of the Sextic Invariant Theory of M21. No DERIVED-LOCKED identity has been falsified across v1.0 $\rightarrow$ v3.0. All CLOSED entries denote former OPEN flags now resolved by post-v2.0 papers.

Glossary Keyword Index (v3.1, expanded with Toolkit terms)

The v3.1 Light Glossary expands the v3.0 index of 102 entries to approximately 130, adding Toolkit terms (T1–T9) and Insight terms (I1–I6) introduced in PARTS VIII and IX, incorporating all post-v2.0 vocabulary. New v3.0 entries are marked ★. Each entry points to the canonical paper and §-anchor for full definition. AI agents navigating Z-Spin Cosmology research should treat this index as the primary disambiguation map between informal phrase-use and DERIVED definition.

Term	Definition / Cross-Reference	Anchor
A (Z-Spin coupling)	Master coupling constant; $A \approx 0.0801\dots$; $A=\rho_X \cdot \rho_Y$ derivation in F9	F1, F9 §3
G (Master Identity)	$G = (Q-Z)$ integer descriptor of cobordism block	F2, M19, M20
Q+Y, Q-Z	Spectral tetrad invariants; Q+Y=17, Q-Z fixes σ -ratios	M20 §3
Z-Spin Mediation	Sector mediation protocol: $X \rightleftharpoons Y$ exchange via Z-handshake	F0-F1, F18
Twin-Reuleaux Pair ★	Geometric realisation of EM duality $V_{XZ} \leftrightarrow R_1, V_{ZY} \leftrightarrow R_2$	S15 §B, F14
Twin-Reuleaux Midpoint ★	Conjugate-decomposed trajectory closing F-F7.11	F14 Strategy B
Reuleaux Cross-Section	$\Pi+\Pi$ half-channel of self-referential equilibrium	F17 §3
Tetrahedral Self-Duality ★	T_d 's E irrep gives $\dim(Z)=2$ with $A=\rho_X \cdot \rho_Y$	F9 §2-3
μ_{Tet} (Tet multiplicity) ★	Dual orientation count =2; origin of factor 2 in $2e^A$	F12 §3
Layer Separation R ★	$R = \mu_{\text{Tet}} \times \text{Hol}_\varepsilon \times \mathcal{P}_{\text{scale}}$ decomposition	F12 §4
Z-Field Master Equation	$A = \rho_X \cdot \rho_Y$ core dynamical identity	F1 §5, F9 §3
Master-of-Master Equation	Composite scaffold across F-M-S spines	F8, F18
FFPP (Five-Fold Parallel Postulate)	Foundational axiom set; v3.0 Sixth Axiom Extension	F8, F15R §3
Five-Axiom Meta-Structure ★	F18 unifying meta-frame; six polarities	F18
Sixth Axiom Extension ★	Stage-Torque Duality lifts FFPP	F15R §3
Bridge Four ★	Commutator Generator; $\dim(Z)=2 \rightarrow \dim(X)=3$ via Lie bracket	F18 §4
Stage-Torque Duality ★	Time-rotation co-definition	F15R §2-3
Discrete-Continuous Resolution ★	Sixth polarity in F18 meta-structure	F18 §6

i-Tetration Internal Time ★	Unifies t_{strobo} , t_{phase} , t_{clock} ; $\Delta v/\Delta n=2A/\pi$	F10
Möbius Chronology ★	Y-frame metric trio; cycle-index unobservability	F13
Y-Hubble Radius ★	$\ell_P \times Y^2(1-2A) \approx 30.23 \ell_P$	F13 §3
Y-Time Dilation	$\exp(\pi/A) \approx 10^{17}$ contraction-frame ratio	A8R §SA.7, F10 §4
Y-frame metric trio ★	Proper time / Y-Hubble / contraction-rate triad	F13 §3
Stroboscopic Step t_{strobo}	Discrete-time origin of Z-Spin clock	F10 §2
Berry-phase clock t_{phase}	Continuous-rotation clock partner	F10 §2
Z-Clock Coordinate t_{clock}	Master clock realised by i-tetration	F10 §3
First Rotation Event ★	$\Theta(0) \cdot T_P = A$; Planck-unit-establishing event	F15R §2
Operational Observer Coordinate (OOC) ★	Self-referential fixed point closing M11 §H16	F11
NON-CLAIM (Six)	Six explicit no-claim statements vs phenomenological consciousness	F11 §6
Self-Referential Fixed Point ★	OOC convergence in F11 §5	F11 §5
Contracting Universe Dynamics ★	A8R 22/22 PASS; three Bridges + $ z^* \approx B$	A8R
Symmetry-Asymmetry Unified View ★	A8R §SA reconciliation programme	A8R §SA
Banach-Tarski Origin ★	$F_2 \rightarrow D_4$ functor; Julia $J(T)$ ZF-analogue	A9R
Two-Branch $(1+A)(1-2A)$ ★	A9R cosmological doubling-halving identity	A9R §3
Three Bridges (A8R) ★	$ z^* \approx B$, η_{topo} formula, Δ via L2	A8R §3
Twin-Reuleaux EM Duality ★	S15 35/35 PASS; Maxwell from handshake	S15
Five Pillars (S15) ★	$V_{XZ} \leftrightarrow R_1$, $V_{ZY} \leftrightarrow R_2$, Maxwell, $SO(3)/SU(2)$, $U(1)$ gauge	S15 §1-5
Spectral Tetrad Sub-Isotype ★	M20 σ -ratios EXACT; pentagon-hexagon $\arctan(2 \cdot \varphi^k)$	M20
Sextic Invariant Theory ★	M21 falsifies 1-loop CW closure for M20.A	M21
Forcing Theorem $((p-1)(q-1)=p)$ ★	M19 unique solution (2,3); $\varphi(Y^2)=G$	M19 §2-3
Arithmetic-Dedekind Scaffold ★	M22 v1.1 60/60 PASS; $K=@(\sqrt{-3}, \sqrt{-11})$	M22
Pillar V scalar-kernel no-go ★	M22 obstruction theorem	M22 Pillar V

Y-Sector RH Contribution Map ★	M23R 31/31 PASS; Four Dragons D4a-D4d	M23R
Four Dragons D4a-D4d ★	M23R Y-sector RH contribution decomposition	M23R §3
Self-Referential Info-Compression Equilibrium ★	F17 four-layer 50/50 (t-Tet, Reuleaux, ME, XOR)	F17
Four-Layer Equivalence ★	F17 polyhedra $\leftrightarrow$ Reuleaux $\leftrightarrow$ Master Eq $\leftrightarrow$ Channel	F17 §3-6
Face Polygon Spectral Zeta ★	M24 Theorem D.1 Legendre archimedean decomposition	M24
Composite Field $K=\mathbb{Q}(\sqrt{-3},\sqrt{-11})$ ★	M22, M25 base arithmetic field	M22, M25
J-Yakoboylu Bridge ★	M25 $\xi_K(s) = (1/4\sqrt{33}) \cdot \xi(s) \cdot \Lambda(\chi_{-3}) \cdot \Lambda(\chi_{-11}) \cdot \Lambda(\chi_{33})$	M25
V_4 -Equivariant ZBSI ★	M26 cobordism-history fiber realization	M26
Three-Wall Map W1/W2/W3 ★	M26 quantitative wall map	M26 §4
Wall W3 Closure ★	M27 Kostant Cubic Dirac BRST closure	M27
Z-Spin RH Bridge ★	M28 30/30 PASS; $\sigma=1/2 \leftrightarrow j=1/2$	M28
Möbius-trace Infinity ★	M28, M30 six-route construction	M28 §3, M30 §2
Z-Funnel Spectral Retraction ★	M29 79/79 PASS; A_5 -Hilb(X_F) CY3	M29
Stapledon Bridge ★	M29 conditional bridge from string compactification	M29 §2
A_5 -Hilb(X_F) CY3 ★	Calabi-Yau threefold; Hodge (5,15)	M29 §3
$R_Z=0$ Conditional ★	M29 conditional residue identity	M29 §5
Twenty-One 1/2 Catalogue ★	M30 enumeration of 1/2-instances across the framework	M30 §3
Three-Layer Duality ★	Sector / Frame / Protocol layered duality	M30 §4
Cross-Coupled Sector Duality (W2) ★	M31 v3.0 Three-in-One reading	M31
Lemma M31.0 Non-Separability ★	M31 W2 non-separability theorem	M31 §0
Z_2 -Parity Selection Rule ★	M31 selection rule on cross-coupling	M31 §3
SDRP (Self-Dual Replication Principle) ★	M31 §4 derivative principle	M31 §4
Spinor-Cycle Averaged Feshbach Criterion ★	M32 56/56 PASS; $R_Z=0$ EXACT (Z-even, 4π -avg)	M32

Path-Reversal Lemma M32.X ★	M32 reversal identity; $\pi/10 \leftrightarrow \pi/5$ doubling	M32
N=20 Spinor Closure ★	M32 spinor-cycle closure index	M32 §4
Path γ -revised Reading C ★	M33 D4a-D4d integrated Tr identity	M33 §C
Two-Protocol Theorem ★	F16 $P_a^{(n)}$ vs $P_b^{(n)}$ Wilson-loop measurement	F16
$P_a^{(n)}, P_b^{(n)}$ ★	Wilson-loop survival probabilities	F16 §2
Cosmos-Human Isomorphism ★	T4 36/36 PASS; (Body, DNA, Brain) decomposition	T4
Six-Step Telomere Hierarchy ★	T4 life-cycle map; Hayflick limit $\approx 2\pi/A \approx 78.45$	T4 §3-6
(Body, DNA, Brain) Decomposition ★	T4 sector decomposition	T4 §2
NC-4 ★	Translational NO physical realization claim	T4 §6
Bootstrap Chain $B0 \rightarrow A$	Foundational chain for Z-Spin emergence	F0, M11, M19
DERIVED-LOCKED	Epistemic tag for proven, verification-passed identities	Legend
DERIVED-CONDITIONAL	Holds modulo a stated conditional; awaits unconditional proof	Legend, M22, M27
DERIVED-CANDIDATE ★	Candidate derivation pending one closure step	Legend (v3.0 expanded)
DERIVED-interpretation ★	Interpretive derivation tag (v3.0)	Legend (v3.0 expanded)
TESTABLE-LONG ★	Long-horizon testable prediction tag	Legend, T4 §6
BOOTSTRAP-PROGRAMME ★	Programmatic bootstrap-status tag	Legend (v3.0 expanded)
NOT EVALUATED	Pre-verification status	Legend
Z-Spin Cosmology	The umbrella programme name	F0
Cobordism-History Fiber	BV-BFV target structure for V_4 -equivariant content	M26, M27, M33
BV-BFV Scaffold ★	Cobordism-history quantization framework	M26 §1, M33
BRST Closure	Quantum gauge-symmetry closure programme	M22, M27
Cubic Dirac Operator (Kostant)	Exterior-algebra Dirac on H-spaces	M27 §3
Hayflick Limit	Cellular replication limit; T4 maps to $N_{(2\pi)}=2\pi/A$	T4 §6

Hodge (5,15)	Hodge numbers of A_5 -Hilb(X_F) CY3	M29 §3
Pentagon-Hexagon Stabilizer ★	M21 stabilizer group structure	M21 §3
Phenomenological Observer	Informal/phenomenal observer; F11 NON-CLAIM target	F11 §6
Polyhedral Integers	Integer-valued descriptors of polyhedral spectra	M19, M20
Master Equation (channel layer)	F17 information-theoretic equilibrium	F17 §5
XOR Channel Layer	F17 fourth equivalence layer	F17 §6
Z-Sector Wilson Loop	Holonomy probe of Z-mediation	F16
Frame Equivalence ★	F13 sectoral frame equivalence statement	F13 §4
Cycle-Index Unobservability ★	F13 fundamental unobservability theorem	F13 §5
Master Falsification Registry	Programme-wide falsifiability ledger (v3.0)	v3.0 §MFR
Source Paper Map	ZS-XX $\rightarrow$ §Y.Z [N/N PASS] index	Part 0
Verification Summary	Cumulative N/N PASS statistics	Title block
Light Edition	Cross-paper integrator format for AI navigation	v3.0 Purpose
Lighthouse Function ★	Programme-navigation aid for AI agents (v3.0)	v3.0 Purpose
T1 Five-Axiom Diagnostic Lattice ★	v3.1 Tool: identifies failing axiom in stuck equations	VIII §1, F18 §3, §6
T2 Bridge Four (Commutator Generator) ★	v3.1 Tool: $\dim(X)=3$ from $\dim(Z)=2$ via Lie commutator	VIII §2, F18 §6, §15 §3
T3 Hermitian-to-PK Switch ★	v3.1 Tool: re-read Feshbach sandwich under $R_Z^{\wedge} ZS$ not $R_Z^{\wedge} H$	VIII §3, M32 §2-3
T4 Two-Layer Phase Quantum ★	v3.1 Tool: separate amplitude ($\pi/10$) from operator ($\pi/5$) layers	VIII §4, M32 §4
T5 N=20 Spinor-Cycle Averaging ★	v3.1 Tool: 4π physical closure (not N=10 algebraic minimum)	VIII §5, M32 §5
T6 Three-Way Split of OPEN flags ★	v3.1 Tool: $A^{\wedge} \text{closeable}$ / $B^{\wedge} \text{one-gap}$ / $C^{\wedge} \text{inherited}$ decomposition	VIII §6, M32 §6
T7 Cross-Coupling Discipline ★	v3.1 Tool: Lemma M31.0 — sum-form fails, bilinear required	VIII §7, M31 §4.0
T8 Π_Z Projector Routing ★	v3.1 Tool: route multiple sub-targets through $\Pi_Z=(1/2)(I+J_Z)$	VIII §8, M33 Reading C

T9 OOC Self-Referential Closure ★	v3.1 Tool: replace phenomenological agent with corpus-internal fixed point	VIII §9, F11 §5-6
I1 Symmetry-Asymmetry Unified View ★	v3.1 Lens: deviation-from-(1, 1/2) reading; $(1+A)(1-2A) \approx 0.9071$	IX §1, A8R §SA
I2 Three-Layer Duality ★	v3.1 Lens: Sector / Frame / Protocol layered reading	IX §2, M30 §6
I3 Six-Route Möbius-Trace Infinity ★	v3.1 Lens: external count-infinity $\leftrightarrow$ internal closure	IX §3, M30 §3
I4 Twenty-One $\frac{1}{2}$ Catalogue ★	v3.1 Lens: $21 \dim(Z)=2$ manifestations + 3 unity = 24	IX §4, M30 §3.5
I5 Four-Layer 50/50 Equivalence ★	v3.1 Lens: t-Tet / Reuleaux / Master Eq / XOR same equilibrium	IX §5, F17 Theorem 5.1
I6 Banach-Tarski Origin ★	v3.1 Lens: $F_2 \rightarrow D_4$ amenability functor; Julia J(T) ZF-analogue	IX §6, A9R
Master Bypass Index ★	v3.1: 16-symptom $\rightarrow$ tool/insight lookup table	after PART IX
Solution Toolkit ★	v3.1: PART VIII operational tools T1–T9 (this volume)	PART VIII
Insight Gallery ★	v3.1: PART IX re-interpretation lenses I1–I6 (this volume)	PART IX
Z-mediator projector Π_Z ★	$(1/2)(I+J_Z)$; idempotent projector onto J_Z -EVEN	M31 §4.0 Def M31.0c
Path-Reversal residual $R_Z^{\wedge ZS}$ ★	$B \cdot Q^{-1} \cdot B^{\wedge PK}$; distinct from Hermitian $R_Z^{\wedge H}$	M32 §2
Reading C ★	M33 Π_Z -routed Tr identity for D4a-D4d	M33 §C
Frame Equivalence (A8R)	X-observer $\leftrightarrow$ Y-observer; HYPOTHESIS-strong	A8R §SA.4

Total v3.0 Glossary entries: 102. ★ marks post-v2.0 vocabulary. Entries without ★ are preserved from v2.0 Light with definitions reaffirmed against the new v3.0 corpus. No v2.0 definition has been retracted.

References (v3.1, preserved from v3.0)

The v3.0 corpus introduces additional external mathematical scaffolding. The references below are imports cited inside the post-v2.0 papers; v2.0 references are preserved in their respective source papers and are not duplicated here.

Internal references (v3.0 corpus, papers ZS-72 through ZS-101)

- ZS-M18 v1.0 — Free-Exploration Session Log: Speculative Prime-Polyhedral Correspondences. 51/51 PASS.
- ZS-A8 v1.0 (Revised) — Contracting Universe Dynamics: The Polyhedral-Tetration Bridge. 22/22 PASS.
- ZS-S15 v1.0 — Twin-Reuleaux Pair as Geometric Realization of EM Field Duality. 35/35 PASS.
- ZS-M19 v1.0 — Number-Theoretic Scaffold of Z-Spin Polyhedral Integers. 68/68 PASS.

- ZS-A9 v1.0 (Revised) — Banach-Tarski Origin of Cosmological Doubling-Halving Symmetry. 47/47 PASS.
- ZS-F9 v1.0 — Tetrahedral Self-Duality and the Hexagonal Mediation Structure. 44/44 PASS.
- ZS-F10 v1.0 — i-Tetration Internal Time: Stroboscopic Step / Berry Phase / Z-Clock unification. 30/30 PASS.
- ZS-F11 v1.0 — Operational Observer Coordinate and the Self-Referential Fixed Point. 38/38 PASS.
- ZS-F12 v1.0 — Tetrahedral Dual Orientation Multiplicity and Layer Separation. 20/20 PASS.
- ZS-F13 v1.0 — The Möbius Chronology: Unified Cyclic Timeline. 24/24 PASS.
- ZS-F14 v1.0 — Kepler-Lyapunov Conjugate Decomposition of the Twin-Reuleaux Midpoint. 42/42 PASS.
- ZS-F15 v1.0 (Revised) — Time-Rotation Co-Definition: First Rotation's Planck-Unit-Establishing Event. 35/35 PASS.
- ZS-M20 v1.0 — Spectral Tetrad Sub-Isotype Assignment Theorem. 50/50 PASS.
- ZS-M21 v1.0 — Sextic Invariant Theory of the Icosahedral Yukawa Tensor. 33/33 PASS.
- ZS-F16 v1.0 — The Two-Protocol Theorem for Z-Sector Wilson Loop Measurement. 24/24 PASS.
- ZS-M22 v1.1 — Five-Pillar Arithmetic-Dedekind Scaffold. 60/60 PASS.
- ZS-M23 v1.0 (Revised) — Y-Sector RH Contribution Map. 31/31 PASS.
- ZS-F17 v1.0 — Self-Referential Information-Compression Equilibrium Theorem (Four-Layer). 24/24 PASS.
- ZS-M24 v1.0 — Face Polygon Spectral Zeta and Archimedean Completion. 35/35 PASS.
- ZS-M25 v1.0 — Composite-Field Archimedean Completion and J-Twisted Yakaboylu Bridge. 26/26 PASS.
- ZS-M26 v1.0 — V_4 -Equivariant ZBSI on the Cobordism-History Fiber. 24/24 PASS.
- ZS-M27 v1.0 — V_4 -Equivariant Cobordism BRST Closure via Kostant Cubic Dirac Operator. 24/24 PASS.
- ZS-M28 v1.0 — The Z-Spin RH Bridge. 30/30 PASS.
- ZS-M29 v1.0 — Z-Funnel Spectral Retraction with Stapledon Bridge. 79/79 PASS.
- ZS-M30 v1.0 — Z-Spin Duality and the RH Bridge — Möbius-Trace Infinity, Two-Face Mirroring, $21 \cdot \frac{1}{2}$ Catalogue. 32/32 PASS.
- ZS-M31 v3.0 — W2 as Cross-Coupled Sector Duality: Three-in-One Reading. 36/36 PASS.
- ZS-M32 v1.0 — A Spinor-Cycle Averaged Residual Criterion for Feshbach Closure. 56/56 PASS.
- ZS-F18 v2.1 — The Five-Axiom Meta-Structure. 36/36 entry checks PASS.
- ZS-M33 v1.0 — V_4 -Equivariant Weil Functional Closure via Path γ -revised Z-Mediator Reading. 52/52 PASS.
- ZS-T4 v1.0 — The Cosmos-Human Isomorphism: Six-Step Telomere Hierarchy. 36/36 PASS.

External imports (post-v2.0)

- Connes, A., Consani, C., Moscovici, H. (2024) — Spectral framework for the Riemann zeta archimedean factor. [imported by M24, M28, M30]
- Burnol, J.-F. (2002, 2004) — Two-sided spectral interpretation of zeta zeros and Möbius traces. [imported by M28, M30]
- Świerczkowski, S. (1958) — Free non-abelian decomposition F_2 acting on S^2 . [imported by A9R]
- Stapledon, A. (2010) — Stringy Hodge numbers of toric varieties; conditional bridge framework. [imported by M29]
- Kostant, B. (1999) — Cubic Dirac operator on equal-rank pairs and Lie-algebra cohomology. [imported by M27]

- Alekseev, A., Mnev, P. (2018) — One-dimensional Chern-Simons / BV-BFV cobordism quantization. [imported by M26, M27, M33]
- Yakaboylu, E. (cited via J-twisted bridge in M25) — Twisted L-function bridges for composite imaginary fields.
- Standard arithmetic and group-theoretic references (Iwasawa, Tate, Weil, Deligne) are inherited from v2.0 and are not re-listed here.

All external imports are used as black-box scaffolding. No external result has been modified by the Z-Spin Cosmology corpus; usage is limited to citation, instantiation, and bridge construction.

Version History

Version	Year	Scope of Update
v1.0 Light	—	Initial Light edition ($\leq$ ZS-F0...F8 + M1...M11)
v2.0 Light	—	90-paper integration; 376/376 PART VI PASS; v2.0 Glossary 88 entries
v3.0 Light	2026	30-paper extension (ZS-72 to ZS-101); ~1,800+ cumulative PASS; Glossary 110 entries; new Master Falsification Registry; PART IV/V/VII expansions; Bridge Four (Commutator Generator); Möbius Chronology; J-Yakaboylu Bridge; V_4 -equivariant BRST closure; $21 \cdot \frac{1}{2}$ Catalogue; Cosmos-Human Isomorphism.
v3.1 Light	May 2026	Solution-Toolkit upgrade. Adds PART VIII (9 methodological tools T1-T9 distilled from F18, M32, M31, M33, F11), PART IX (6 re-interpretation lenses I1-I6 from A8R, M30, F17, A9R), and Master Bypass Index (16-symptom lookup table). v3.0 lighthouse-of-pointers promoted to a working solution kit. Glossary, Falsification Registry, References preserved verbatim from v3.0.

Major v3.0 closures of v2.0 OPEN flags:

- F-F7.11 [CLOSED] — closed by ZS-F14 Strategy B (Conjugate Uniqueness).
- M11 §H16 [CLOSED] — closed by ZS-F11 Operational Observer Coordinate.
- M22 W3 [PROMOTED to DERIVED-CONDITIONAL] — via ZS-M27 Kostant Cubic Dirac BRST closure.
- M22 W2 [PROMOTED to DERIVED] — via ZS-M31 v3.0 Cross-Coupled Sector Duality.
- Master-Equation Sixth-Axiom gap — addressed by ZS-F15R Stage-Torque Duality.

- Phenomenological observer ambiguity — settled by ZS-F11 §6 NON-CLAIM (Six).
- Pentagon-Hexagon angle conjecture — proved EXACT by ZS-M20 ($\arctan(2 \cdot \phi^k)$).
- 1-loop CW closure for M20.A — formally FALSIFIED by ZS-M21 Sextic Invariant Theory.

— End of The Book of Z-Spin Cosmology v3.1 — Light Edition —

Compiled by: Claude (cross-paper integrator), under Kenny Kang's editorial direction. v3.0 — May 2026.
Lighthouse function: navigation aid for AI agents engaging with the Z-Spin Cosmology research programme. No claim of physical realisation is made by this Light edition; canonical statements live in the source papers ZS-F0 through ZS-T4.

